



FORMULA STUDENT VEHICLE DYNAMICS

THE COMPLETE HANDBOOK

Tyres • Suspension • Steering • Telemetry • Build & Validation

TEAM BLITZ | MIST

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Design → Fabricate → Test → Validate • Revision 1

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How to use this handbook

This handbook is the complete vehicle-dynamics reference for Team Blitz — everything needed to take a Formula Student car from a blank sheet to a competition-ready, validated machine, written so a new member can start with no prior coursework and finish able to design, build and analyse the VD package. It brings together three previously separate books plus a new practical section, as four parts:

- **Book One — Tyres:** the tyre — slip, grip, load sensitivity, the Magic Formula, and the FSAE Tire Test Consortium. The force the whole car is built to deliver.
- **Book Two — Vehicle Dynamics & Suspension:** mass, load transfer, balance, suspension geometry, springs, dampers, steering, how every decision couples, and telemetry. The machine that delivers load to the tyres.
- **Book Three — The Build Playbook:** the step-by-step execution plan — ten phases, decision gates, procurement timing, and the dependency map of what to fix before what.
- **Book Four — Practical & Reference:** fabrication and workshop skills, project management and budget, competition-day operations, and a ready-to-use templates pack.

How to read it

Read Book One first; the rest assume its vocabulary. Then Book Two for the understanding, Book Three to execute, and Book Four for the hands-on and the templates. The merged glossary, symbol table and reading list at the back serve all four.

Conventions

- **Bold** — a term defined on first use; grey “where...” lines define every symbol in a formula.
- **Boxes** — red-edged boxes are the ideas that win points and lap time; amber boxes are STOP-gates; rust boxes flag when to order parts; green/red tables are the hard practical rules.
- **Links** — this is a hyperlinked PDF. The contents list is clickable, and every cross-reference (“**Book Two Part E**”, “**Phase 3**”, “**Appendix A**”) is a live link — click to jump, then use your reader’s back button to return.

The one idea that ties the whole handbook together

The tyres make the grip; the car decides how much of it you get and how it splits front-to-rear; the data proves it. Tyre → vehicle dynamics → build → measurement. Keep that chain in mind and every part of this book finds its place.

BOOK ONE

FORMULA STUDENT VEHICLE DYNAMICS

Tyres

The force everything else is built to deliver — slip, grip, load sensitivity, the Magic Formula, and the FSAE Tire Test Consortium.

Part I — Why the tyre is the whole game

Four patches of rubber, each about the size of your palm, are the only places your car touches the ground. Every force that accelerates, brakes, or turns the car passes through those four contact patches. The engine, the aero, the chassis stiffness — all of it exists only to put load into the tyres in the right place at the right time. This is why the TTC literally says “the tyres are the most important vehicle component,” and why a good tyre engineer is worth more points than a good anything-else-engineer.

The contact patch. The

If you could film the road through a glass track, you would see each tyre pressing a roughly oval footprint onto the surface — the contact patch (also “footprint”). Its area is approximately the vertical load on the tyre divided by the contact pressure, which is close to the inflation pressure. That single sentence already contains three design levers: load (mass, weight transfer, downforce), inflation pressure, and tyre size (which sets how that area is shaped). Hold onto it.

A Formula Student car weighs roughly 230–320 kg with driver and makes something like 1.4–1.8 g of grip. That means each tyre is routinely asked for lateral forces comparable to the entire weight it carries. The tyre does this not by “gripping” like glue but by deforming and sliding in a controlled way — which is the subject of this whole guide.

The mental model that separates beginners from engineers

Beginners think of grip as a fixed number (“this tyre has grip X”). Engineers think of grip as a surface: force depends on slip, load, camber, pressure, temperature and speed all at once. Your job is not to “find the grip” — it is to characterise that surface from data and then design the car so all four tyres live on the good part of it.

Part II — Anatomy, construction and how to read a tyre size

What a tyre is made of

A tyre is a pressure vessel and a structure at the same time. From the inside out:

- **Inner liner** — the airtight inner layer that holds the inflation gas; the tyre is really the air plus the casing working together.
- **Carcass / plies (the casing)** — layers of cords (textile or steel) set in rubber that give the tyre its shape and carry the inflation and road loads. The angle of these cords defines the two big families below.
- **Beads** — stiff hoops of steel wire that lock the tyre onto the wheel rim and react all the forces into the wheel.
- **Sidewall** — the flexible side of the tyre between bead and tread; its height and stiffness set a lot of the spring and camber behaviour.
- **Belts** — reinforcing layers (often at a shallow angle, sometimes steel or aramid) under the tread that stabilise the footprint. Competition slicks may use a simpler belt package than road tyres.
- **Tread and compound** — the rubber that actually touches the road. Its chemistry is the compound; its pattern (or lack of one, on a slick) is the tread pattern.

Bias vs radial — the one construction distinction you must know

The ply angle defines two constructions with genuinely different behaviour:

- **Bias-ply (cross-ply)** — cords run diagonally (bias) from bead to bead, criss-crossing. The sidewall and tread are coupled, so the tyre is stiffer in a “chunky” way, warms up fast, and is forgiving (broad, gentle force peak). Most FSAE competition tyres — Hoosier, many Goodyear — are bias-ply, which is exactly why they suit amateur drivers and short events.
- **Radial** — cords run straight across (radially, 90° to the direction of travel) with separate stiff belts. Sidewall and tread act more independently: lower rolling resistance, more stable at high speed, usually a sharper force peak. Almost all road car tyres are radial; some FSAE tyres (e.g. certain Continental and Avon offerings) are radial.

Why this shows up in reports

A compound often praised for “providing ~99% of peak lateral force over an 8° slip-angle range” and being “driveable ... beneficial for amateur drivers” is describing a broad, forgiving peak — a hallmark of a softer bias-ply slick. That broad peak is a feature for a student driver who cannot feel a 0.2° slip-angle error. You will make the same trade in your own car.

How to read a tyre size

FSAE tyres are quoted in two systems. Learn to translate both instantly:

Table II-1. Tyre size and compound nomenclature. OD = overall (outer) diameter.

Marking	Means	Example from the reports
18 × 6.0 – 10	OD × section width – rim, all in inches	Hoosier 18.0×6.0-10 R20 (a common data proxy)
18 × 7.7 – 10	18" tall, 7.7" wide tread, 10" rim	a custom 18×7.7-10

Marking	Means	Example from the reports
6 / 18 – 10	width / OD – rim (same numbers, order swapped)	e.g. “Hoosier 6/18-10 LC0”
R20, LC0	compound code (maker-specific)	R20 and LC0 are Hoosier compounds

Read it like this: So “Hoosier 6/18-10 LC0” and “Hoosier 18.0×6.0-10” are the same size written two ways; only the compound differs.

Two sizing facts drive design choices you will have to defend:

- **Smaller diameter / rim** — a shorter tyre has less rotational inertia and less unsprung mass (both good for how fast the wheel follows the road), and lowers the car. This is why many FSAE teams move from 13" to 10" wheels. The cost is a smaller, more pressure-sensitive contact patch.
- **Wider section** — a wider tyre gives a bigger contact patch and usually higher peak lateral force, but adds mass, rotational inertia, aero drag and sometimes more load sensitivity. A wider tyre — say 7.7" against a 6.0" Hoosier used as a data proxy — means you must remember the width gap when you borrow the proxy's numbers.

Part III — The language: axis systems, forces, moments and slip

Before any numbers mean anything, you need the coordinate system every book and dataset uses. Get this once and the rest of the field opens up.

The SAE tyre axis system

Picture the contact patch as the origin. The standard (SAE / ISO) tyre axes are:

- **x** — points forward, in the direction the wheel is heading.
- **y** — points to the right (SAE) — the “lateral” direction.
- **z** — points down into the road (SAE), so vertical load is a positive number.

Each axis carries a force and a moment (a twisting effort about that axis):

Table III-1. The six tyre forces and moments. You will mostly live in F_y , F_z and M_z .

Symbol	Name	Plain meaning
F_x	Longitudinal force	Drive (+) or brake (–) force along the direction of travel
F_y	Lateral force	Cornering force, sideways — what turns the car
F_z	Vertical load (normal load)	How hard the tyre is pressed onto the road
M_x	Overturning moment	Tyre trying to roll over on its sidewall
M_y	Rolling resistance moment	Resists the wheel’s rotation (losses)
M_z	Aligning moment (self-aligning torque)	Twist about the vertical axis that tries to straighten the steering — your steering “feel”

Sign-convention warning (this trips up everyone, including pros)

SAE has z pointing down; ISO has z pointing up, which flips the sign of several quantities. TTC raw data is in one convention and many tools silently convert. When a lateral-force plot looks upside-down or a camber effect has the “wrong” sign, it is almost always an axis-system mismatch, not a physics error. Always state which convention your tool uses — and check it against a case you understand (e.g. positive slip angle should make the tyre push the car toward the inside of the corner).

Slip angle — the master variable for cornering

A rolling tyre does not point exactly where it travels. The small angle between the direction the wheel is pointing and the direction the contact patch is actually moving is the slip angle, α (alpha). That mismatch distorts the tread rubber sideways; the distorted rubber pushes back; that push is the lateral force F_y . No slip angle, no cornering force.

$$\alpha = \arctan(V_y / V_x)$$

where α = slip angle; V_y = sideways speed of the wheel centre; V_x = forward speed. Small angles: $\alpha \approx V_y / V_x$ (in radians).

Typical FSAE operating slip angles are a few degrees — the commonly quoted “4–6°” and “8°” bands are exactly this. “Building” or “peaking” slip angle just means where on the F_y -vs- α curve the tyre makes its most force.

Slip ratio — the master variable for traction and braking

The longitudinal equivalent of slip angle is slip ratio, κ (kappa) or SR. It measures how much faster (or slower) the tread surface is moving than the road, because the tyre must deform fore-aft to transmit drive or brake force.

$$\kappa = (\Omega \cdot R_e - V_x) / V_x$$

where κ = slip ratio (often $\times 100$ for %); Ω = wheel spin rate (rad/s); R_e = effective rolling radius; V_x = forward speed. $\kappa > 0$ drive/wheelspin, $\kappa < 0$ braking/lock-up, $\kappa = -1$ fully locked wheel.

Peak traction and braking happen at a modest slip ratio (roughly 0.1–0.2, i.e. 10–20%), not at zero and not at a locked wheel. That is the whole reason ABS and traction control exist: they hold the tyre near that peak. On an FSAE car it is why a good launch feeds just a little wheelspin, and why a locked brake ($\kappa = -1$) actually stops you slower than threshold braking.

Camber (inclination angle)

Camber angle, γ (gamma), is the tyre's lean when you look at the car from the front. Negative camber means the top of the tyre leans in toward the car. A leaned tyre generates a small lateral force all by itself — camber thrust — and, more importantly, keeps the contact patch flat on the road when the car rolls in a corner. Pacejka/ISO call this the inclination angle. More on using it in [Part IX](#).

Part IV — How a tyre actually makes grip (and why μ can exceed 1)

Two grip mechanisms

Rubber friction is not the textbook “block on a ramp” friction. Two mechanisms act together:

- **Adhesion** — rubber molecules momentarily bond to the road surface, stretch, and snap free, over and over. Dominant on smooth, clean, dry surfaces. This is why a clean dry track grips far more than a dusty one even though both are “dry.”
- **Hysteresis (deformation)** — the road’s microscopic bumps push into the rubber; as the tyre slides, the rubber is continuously deformed and springs back late, losing energy and dragging on the surface. Dominant on rough surfaces and the reason a slightly worn, textured surface can grip well.

Because both mechanisms depend on the rubber deforming, grip depends on temperature, speed and how hard you press — friction is not a constant for rubber the way it is for steel on steel.

Why the friction coefficient is greater than 1

The friction coefficient μ (mu) is just lateral (or longitudinal) force divided by vertical load:

$$\mu = F / F_z$$

where μ = friction coefficient; F = F_y (cornering) or F_x (traction/braking); F_z = vertical load.

For a brick on a table μ is below 1. For a racing slick μ is commonly 1.3–1.8 and sometimes higher, because adhesion and hysteresis add grip beyond the simple “load $\times$ friction” picture. Quoted values like “ $\mu_Y \approx 1.82$ ” and “ $\approx 1.6\text{ g}$ ” come straight from this: a μ of ~ 1.6 – 1.8 means ~ 1.6 – 1.8 g of cornering, because at steady state lateral $g \approx \mu$ (plus any aero help).

The single number judges will probe: your scaling factor $k \approx 0.65$

Lab tyre machines (like Calspan’s) run on a special high-grip belt surface, so raw TTC μ is optimistic versus real asphalt. Teams multiply raw grip by a scaling factor (you used $k \approx 0.65$) to get a track-representative number. This is standard practice — but you must be able to say WHERE 0.65 came from (a published range, a prior team’s correlation, or your own skidpad). “Grip scaling factor” is the correct term; “belt-to-track” is jargon for exactly this.

The friction circle / ellipse — preview

A tyre has one budget of grip to spend between cornering and accelerating/braking. Spend it all sideways and you have none left for braking; brake at the limit and you cannot also turn. Drawn out, the combinations you can reach form a circle (or, because tyres are usually a bit stronger one way, an ellipse). We make this quantitative in [Part VII](#); for now, hold the picture.

Part V — Lateral force: slip angle, cornering stiffness, aligning moment, camber

This is the heart of a cornering car and the part of a tyre report that matters most. We will walk the lateral-force curve from the origin out past the peak.

The lateral-force vs slip-angle curve

Plot F_y (up) against slip angle α (across) at a fixed load. You get an S-shaped curve with three regions:

- **Linear region (0 to ~2–3°)** — F_y rises almost in a straight line with α . The tyre is mostly gripping, sliding only at the back of the contact patch. The slope here is the single most useful number about a tyre (next heading).
- **Transitional region (~3–6°)** — the curve bends over as more of the contact patch slides. Force still rises but less per degree.
- **Peak and beyond (~6–10°)** — F_y reaches a maximum (the whole patch is near sliding) and then slowly falls. The slip angle at the maximum is the peak slip angle; the force there divided by load is peak μ .

Read a lateral-force plot correctly

On a lateral-force plot the light-load curves peak around 5–6° but the heavy-load curves keep climbing to ~8–10°. That load-dependence of the peak slip angle is normal and important: the tyre you actually run (heavily loaded on the outside of a corner) peaks later than the lightly loaded one. So prefer “builds through 4–6°” over “peaks at 4–6°” — at the true operating load the peak is further out.

Cornering stiffness — the number to quote

Cornering stiffness, C_α (also “Cornering stiffness”, units N per degree or N per radian), is the slope of the F_y -vs- α curve at zero slip angle — how many newtons of cornering force you get for the first degree of slip.

$$C_\alpha = \partial F_y / \partial \alpha \text{ (evaluated at } \alpha = 0 \text{)}$$

where C_α = cornering stiffness (N/deg or N/rad). Steeper curve = stiffer tyre = sharper turn-in and more responsive car.

Why you care: cornering stiffness, front vs rear, is what sets whether the car understeers or oversteers (Part XIII). It is also where a lumped-parameter vehicle model gets its tyre inputs. When a “cornering stiffness vs load” chart compares candidate sizes, it is comparing exactly this slope — and note that the stiffest tyre at low load is not always the stiffest at the real operating load, which is a legitimate reason not to claim “highest stiffness” outright.

Self-aligning moment and pneumatic trail — where steering feel comes from

The lateral force does not act at the exact centre of the contact patch; it acts a little behind it, because the rear of the patch is doing more of the work. That small rearward offset is the pneumatic trail, t_p . Force $\times$ trail is a torque that tries to straighten the wheel — the self-aligning moment (aligning torque), M_z .

$$M_z \approx - F_y \times (t_p + t_m)$$

where t_p = pneumatic trail (tyre); t_m = mechanical trail (from caster/steering geometry); F_y = lateral force. The product is the torque the driver feels through the wheel.

The practical gold in this: pneumatic trail shrinks as slip angle rises and reaches ~zero right around the force peak — so the steering goes light just before the tyre lets go. That lightening is the main cue a good driver

uses to feel the limit. It is also why over-assisted or very-low-caster steering “numbs” the car. For you: Mz data from the TTC is how you can estimate pneumatic trail and set caster so the wheel still has feel at the limit.

Camber thrust and camber stiffness

A cambered (leaned) tyre generates lateral force even at zero slip angle — camber thrust — because the leaned, distorted contact patch tries to roll in a cone. The sensitivity is camber stiffness, C_γ .

$$F_y \approx C_\alpha \cdot \alpha + C_\gamma \cdot \gamma \quad (\text{small-angle, pure lateral})$$

where C_γ = camber stiffness (N/deg of camber); γ = camber (inclination) angle. Camber thrust adds to slip-angle force — a few percent, but it matters at the limit.

Camber thrust is small compared with slip-angle force (camber stiffness is maybe a tenth of cornering stiffness) but it is free lateral force and, more importantly, the reason you run static negative camber: to keep the patch flat and loaded when the car rolls. Your -2° front static camber is this decision. [Part IX](#) shows how to choose it from data rather than by habit.

Relaxation length — the tyre is not instant

A tyre needs to roll a short distance before the lateral force fully builds after you change slip angle. That distance is the relaxation length, σ (sigma) — typically a few tenths of a metre, i.e. a fraction of a wheel revolution. It is why the car responds to steering with a tiny lag, why “step-steer” tests (which report a “settling time”) measure how quickly the car settles, and why very low-inertia wheels feel sharper.

$$\sigma = C_\alpha / K_y$$

where σ = relaxation length; C_α = cornering stiffness (N/rad); K_y = lateral stiffness of the tyre carcass (N/m). Bigger C_α or softer carcass $\rightarrow$ longer lag.

Part VI — Longitudinal force: slip ratio, traction and braking

Everything from **Part V** has a fore-aft twin. Drive and brake forces come from slip ratio κ exactly as cornering force comes from slip angle α .

The longitudinal-force curve

Plot F_x against slip ratio κ at fixed load and you get the same S-shape: a steep linear rise, a peak at a modest slip ratio (~10–20%), then a decline into full wheelspin (drive) or lock-up (braking). The slope at the origin is the longitudinal (slip) stiffness, C_κ .

$$C_\kappa = \partial F_x / \partial \kappa \quad (\text{at } \kappa = 0)$$

where C_κ = longitudinal slip stiffness (N per unit slip ratio). Analogous to cornering stiffness but for traction/braking.

Peak F_x usually happens at a larger effective slip than peak F_y , and the peak longitudinal μ is often a touch higher than peak lateral μ on a slick. This matters in two places: the launch/acceleration event (how much drive the rear tyres can take before spinning) and braking (**Part XIII** and the brake system's lock-up margin).

Effective rolling radius and why κ is subtle

The effective rolling radius R_e is not simply the tyre's outer radius; a loaded tyre squashes, so it travels less per revolution than a rigid wheel of the same size would. R_e sits between the free radius and the loaded (static) radius. Getting R_e slightly wrong is the most common cause of a slip-ratio curve that looks shifted — which is why TTC processing spends care establishing R_e before fitting F_x .

FSAE reality check

Most FSAE teams characterise lateral behaviour thoroughly and treat longitudinal more coarsely, because the big points (skidpad, autocross, endurance) are cornering-dominated and TTC drive/brake data is noisier. Know the longitudinal story, quote peak drive/brake μ , but spend your deepest analysis on F_y .

Part VII — Combined slip and the friction ellipse

In a real corner the tyre is almost never doing only one job. You trail-brake into the apex (braking + cornering) and feed throttle on exit (traction + cornering). Combined slip is how lateral and longitudinal forces share the one grip budget.

The friction ellipse, quantitatively

If the most lateral force the tyre can make is $F_{y,max}$ and the most longitudinal is $F_{x,max}$, then at any instant the usable combination is bounded by:

$$\left(F_x / F_{x,max} \right)^2 + \left(F_y / F_{y,max} \right)^2 \leq 1$$

where Spend grip on F_x and the F_y you can still make shrinks, and vice-versa. Equality ($= 1$) is the limit — the edge of the ellipse. A circle is the special case $F_{x,max} = F_{y,max}$.

This one inequality explains an enormous amount of driving and setup: why you cannot brake and turn at 100% of each, why a smooth driver who separates braking and turning is slow, why a diff that manages wheelspin on exit (a limited-slip differential) buys lap time, and why trail-braking works. The g-g diagram is literally this ellipse drawn in acceleration units — the envelope of combined lateral and longitudinal g the car can reach.

Tie-in to telemetry

Teams often log “G-G plots” to evaluate drivers. A G-G (or g-g) plot is the friction ellipse measured on track: every point is the car’s combined lateral/longitudinal acceleration at one instant. A driver who fills the ellipse out to its edge, and spends little time in the dead centre, is using the tyres. This is the most direct feedback loop between your tyre work and lap time — learn to read it.

Part VIII — Load sensitivity: the most important idea in car setup

If you learn one thing from this guide that a rival team's tyre lead might not know, make it this. It is the reason half of suspension design exists.

The effect

Push harder on a tyre (more vertical load F_z) and it makes more force — but not proportionally more. The friction coefficient μ falls as load rises. Double the load and you get noticeably less than double the grip. This is load sensitivity (also “load dependency” of grip).

$$\mu(F_z) \text{ decreases as } F_z \text{ increases} \rightarrow \partial\mu/\partial F_z < 0$$

where Equivalently, the lateral force rises less-than-linearly with load. Your Figure 1 “peak μ vs load” curve sloping downward IS load sensitivity; the “ $\Delta\mu$ per +1 kN” bar chart is its slope.

Why it drives the whole car

Now put it on an axle in a corner. Weight transfers from the inside tyre to the outside tyre (**Part XIII**). The outside tyre gains load, the inside loses the same amount. Because of load sensitivity, the outside tyre gains LESS grip than the inside tyre loses. So the pair of tyres — at the same total load — makes less lateral force when the load is split unevenly than when it is shared.

The payoff sentence (say this in the design event)

Two tyres at the mean load make more grip than one heavily-loaded plus one lightly-loaded tyre. Therefore anything that reduces lateral load transfer — wider track, lower centre of gravity, lower mass — raises total grip; and shifting load transfer between the two axles (via roll-stiffness distribution, i.e. springs and anti-roll bars) is how you trade grip front-to-rear to balance the car. A good tyre section states it directly: “load sensitivity ... managed by spring rates, roll-centre heights and track width limiting outer-tyre load transfer.” Now you know exactly why that sentence is true.

Two consequences you will use

- **Tyre choice** — a tyre with LESS load sensitivity (flatter μ -vs- F_z) is more tolerant of weight transfer and of a heavy car. When you chose the 18×6 Hoosier proxy and noted its “higher load sensitivity” as the trade-off, this is the penalty you accepted for its higher peak μ — and the reason you then argue you manage load transfer to cash in the peak μ anyway.
- **Balance** — because the heavier axle is penalised more by load sensitivity, a rear-heavy car (yours is 45/55) wants proportionally more of its roll stiffness handled so the rear is not over-transferred — which is why a rear-heavy car runs “the rear roll centre higher than the front to counteract the rear-heavy weight distribution.”

A worked feel for the numbers

Suppose at 600 N a tyre makes $\mu = 1.75$ (1050 N) and at 1200 N it makes $\mu = 1.55$ (1860 N). Share 1800 N evenly (900 N each): roughly $\mu \approx 1.65 \rightarrow \sim 1485$ N each $\rightarrow 2970$ N for the pair. Now split it 300 N / 1500 N: the light tyre $\sim \mu 1.82 \rightarrow 546$ N, the heavy $\sim \mu 1.48 \rightarrow 2220$ N $\rightarrow 2766$ N for the pair. Same 1800 N total,

but ~7% less cornering force when the load is split. That 7% is lap time, and it is created entirely by load sensitivity.

Part IX — The three knobs: pressure, temperature, camber

These are the levers you actually turn at the track. Each one moves the whole force surface, so each is a tool and a trap.

Inflation pressure

Pressure sets the size and shape of the contact patch and the stiffness of the air spring. There is an optimum, and it is lower than most beginners guess for slicks.

- **Too high** — smaller contact patch, higher local pressure in the patch, stiffer and more responsive (higher cornering stiffness up to a point), less rolling resistance — but beyond the optimum the centre of the tread carries too much load, overheats, and peak μ falls. The patch also gets “pointy,” reducing area.
- **Too low** — bigger patch and often more peak grip — up to a point — but the tyre deflects more, the shoulders overheat, the carcass works harder (heat and wear), response gets vague, and rolling resistance climbs. Too low and the tread can even lift in the middle.

FSAE slicks typically run roughly 55–97 kPa (8–14 psi) — a ~69 kPa baseline sits right in this band. The honest way to state it is “highest grip within the tested pressure range,” because grip rising as pressure falls only holds down to the lowest pressure you have data for; extrapolating below that is a trap.

Hot vs cold pressure — the gas-law detail that catches teams out

You set pressures cold, but the tyre heats up and the pressure rises. What matters for grip is the hot (running) pressure. Use the ideal-gas relation to target it:

$$P_{\text{hot}} / P_{\text{cold}} \approx T_{\text{hot}} / T_{\text{cold}} \text{ (absolute temperatures, in kelvin)}$$

where Add 273 to °C before using. Example: 69 kPa gauge $\approx$ 170 kPa absolute cold at 25°C (298 K); at 60°C (333 K) it rises to ~190 kPa absolute $\approx$ 89 kPa gauge. So a ~20 kPa hot rise is normal — set cold low to land on target hot.

Temperature

Every compound has a temperature window where grip peaks. Below it the rubber is glassy and slides; above it the rubber goes greasy and degrades. FSAE events are short, so a compound that reaches its window fast is worth real points on a cold skidpad — which is precisely the argument for a fast-warming compound (one with “faster tyre temperature build”).

You read temperature three ways: a tyre pyrometer (a needle probe) jabbed across the tread right after a run, an infrared surface sensor logged live on the car, or by how the tyre looks and feels. The across-the-tread spread (inside vs middle vs outside) is a camber and pressure diagnostic:

Table IX-1. Tyre pyrometry quick-diagnosis. Take three readings (inner/middle/outer) within seconds of coming in.

Across-tread temperature pattern	What it usually means	Action
Inside edge much hotter than outside	Too much negative camber (or too low pressure)	Reduce static negative camber a touch
Outside edge hotter	Not enough negative camber for the roll you have	Add negative camber
Middle hotter than both edges	Over-inflated (crowned patch)	Lower pressure
Edges hotter than middle	Under-inflated (patch collapsing in centre)	Raise pressure
Even across the tread	Camber and pressure about right	Leave it; record the settings

Camber (using it, not just defining it)

Static negative camber is a compromise. You want $\sim 0^\circ$ of camber at the contact patch when the tyre is most loaded — i.e. on the outside wheel at mid-corner, after the car has rolled and the suspension has moved the wheel through its camber curve. Because the car rolls and the geometry adds camber as the wheel moves (camber gain), you start with static negative camber so you arrive near zero (or slightly negative) under load.

$$\gamma_{\text{dynamic (outer)}} \approx \gamma_{\text{static}} + (\text{camber gain from roll \& bump}) - (\text{body roll angle})$$

where The target is a small negative γ_{dynamic} on the loaded outer tyre so the patch is flat-to-slightly-loaded on its inner edge. Pyrometry (Table IX-1) tells you whether you got it right.

So -2° static front is not a magic number — it is the value that, for your roll stiffness and camber curve, should put the loaded front tyre near its happy camber. If your inner edges run hot, you have too much; if outer edges run hot, too little. This is a measure-and-iterate loop, and exactly the kind of data-driven story the design event rewards.

Part X — Tyre models: the brush model and the Magic Formula

A “tyre model” is a set of equations that reproduces the force surface so a lap-sim or a design study can ask “how much F_y at this α , F_z , γ ?” without a physical test. Two models matter to you: the brush model (for understanding) and the Magic Formula (for doing).

The brush model — understand the shape, do not fit it

Imagine the tread as thousands of tiny independent bristles. As the tyre rolls through a slip angle, each bristle enters the contact patch upright, gets dragged sideways as it travels through the patch, and deflects more the further back it is — until its sideways force exceeds what friction can hold, and it slides. So the front of the patch grips (adhesion region) and the rear slides (sliding region).

- **Low slip** — at small slip almost the whole patch grips → force rises linearly → this region’s slope IS the cornering stiffness.
- **More slip** — the sliding region grows forward from the rear; force keeps rising but more slowly.
- **At the limit** — the whole patch slides → force peaks at roughly $\mu \cdot F_z$ and then eases off.

The brush model also explains pneumatic trail falling to zero at the peak (as the grip region shrinks, the force centroid moves forward) and why combined slip shares one budget (a bristle sliding fore-aft has less left for sideways). You will not fit the brush model to data; you keep it in your head so the Magic Formula’s coefficients mean something. (Full treatment: Pacejka Ch. 3; Milliken RCVD Ch. 2.)

The Magic Formula (Pacejka) — the industry-standard curve you actually fit

The Magic Formula (MF) is a semi-empirical model: it is not derived from physics but is a cleverly shaped equation that fits measured tyre data extremely well with a handful of coefficients, each controlling one visual feature of the curve. “Semi-empirical” is the key phrase — it is curve-fitting with physically meaningful knobs. It is what OptimumTire, MFeval, MF-Tyre and your own tool all use.

The core pure-slip form (for F_y vs α , or F_x vs κ) is:

$$Y(x) = D \cdot \sin\left(C \cdot \arctan\left(B \cdot x - E \cdot \left(B \cdot x - \arctan(B \cdot x)\right)\right)\right)$$

where Y = output force (F_y or F_x); x = input slip (α or κ), usually with a small horizontal shift S_h added; a vertical shift S_v may be added to Y . B , C , D , E are the four shape coefficients below.

Each coefficient owns one feature of the curve — this is why the MF is teachable:

Table X-1. The Magic Formula coefficients. The combination $B \cdot C \cdot D$ is the slope at the origin.

Coeff.	Name	What it controls	Tie to things you already know
D	Peak factor	The maximum force (height of the curve)	$D = \mu_{\text{peak}} \times F_z$. This is your peak grip.
C	Shape factor	How far the curve falls after the peak	≈ 1.3 for F_y , ≈ 1.65 for F_x , ≈ 2.4 for M_z
B	Stiffness factor	How steeply force rises at the origin	Chosen so that $B \cdot C \cdot D$ = the cornering stiffness C_α
E	Curvature factor	The exact shape around the peak	Fine-tunes where/how sharply the peak occurs

$$C\alpha = B \cdot C \cdot D \text{ (slope at the origin = cornering stiffness)}$$

where This identity is why MF is convenient: fit D from the peak, set the product $B \cdot C \cdot D$ to the measured initial slope, and C, E to the post-peak shape.

Real tyres are not the same at every load, camber or pressure, so each coefficient is itself written as a function of vertical load and inclination angle (e.g. the peak μ falls with load — load sensitivity — so D has load terms; camber shifts and skews the curve, so B, S_h , S_v carry camber terms). A full MF tyre file (the “.TIR” file) is just a long list of these sub-coefficients, named p-Dy1, p-Ky1, and so on. You do not memorise them; you fit them to TTC data and sanity-check the handful that have physical meaning (peak μ , cornering stiffness, peak slip angle).

Scaling factors — and the grip scaling factor $k \approx 0.65$

MF includes a set of scaling factors (the Greek λ terms, “lambda”) that multiply whole effects so you can stretch a fitted tyre to a new condition without refitting. The most important for you is the peak-friction scaling $\lambda\mu$:

$$Fy_{track} = \lambda\mu \cdot Fy_{lab} \text{ with } \lambda\mu \approx 0.65$$

where $\lambda\mu$ (lambda-mu) = friction scaling factor. Setting $\lambda\mu \approx 0.65$ is the formal, defensible way to apply your belt-to-track grip scaling inside the model — far better than hand-multiplying the output.

Say it the professional way

Instead of “we multiplied grip by 0.65,” say “we applied a friction scaling factor $\lambda\mu \approx 0.65$ to the Magic-Formula peak to make the lab data track-representative.” Same action, but it signals you know the model. Then be ready to justify 0.65 from a source or your own skidpad correlation.

Model versions (so the names stop being scary)

- **MF 5.2** — a widely used pure+combined-slip version; MFeval and many FSAE tools fit this to TTC data. Great default.
- **MF 6.1 / 6.2** — adds turn-slip and better large-camber/low-speed behaviour; the modern standard. 6.2 extends it further.
- **MF-Tyre / MF-Swift** — a commercial Siemens/TNO implementation you will see referenced; MF-Swift adds ride/transient dynamics. You do not need the paid tools for FSAE.

Part XI — The FSAE Tire Test Consortium (TTC) and the tools

You cannot fit a Magic Formula without data, and you almost certainly cannot afford to rent Calspan yourself. The TTC exists precisely to solve that for student teams — it is the single most valuable resource in your corner of the car, and knowing it cold is part of being the tyre person.

What the TTC is

The Formula SAE Tire Test Consortium (TTC), founded in 2005 and run with Calspan, pools money from member teams to run professional tyre tests and shares the raw data with all members. For a one-time fee (on the order of a few hundred US dollars — about USD 500 historically) a team gets access to every completed test round's data. Tests run on Calspan's Tire Research Facility (TIRF) flat-belt machine at roughly 25 mph, chosen to match average FSAE speeds.

- **How to get it** — access is via the member forum (fsaettc.org); your university registers once and can then download every round.
- **What is in it** — successive rounds have covered common FSAE sizes and compounds from Hoosier, Goodyear, Continental, Avon and others — including the Hoosier 18×6-10 family often used as a data proxy.
- **Action for you** — if Team Blitz is not already a member, this is the highest-leverage ~USD 500 the team can spend. Check the public members list first — your school may already have access from a previous year.

What a test round contains

Each round sweeps the tyre through conditions on the belt and logs dozens of channels. The ones you live in:

Table XI-1. Core TTC data channels. A single run file sweeps SA while holding FZ, IA and P, then steps to the next combination.

Channel	Symbol	Why it matters
Slip angle	SA / α	x-axis of the lateral-force curve
Slip ratio	SR / κ	x-axis of the drive/brake curve
Vertical load	FZ	the load the tyre is pressed at (they sweep several)
Inclination angle	IA / γ	camber (they sweep a few values)
Lateral force	FY	the cornering force you fit
Longitudinal force	FX	traction/braking force
Aligning moment	MZ	self-aligning torque → pneumatic trail, feel
Pressure	P	they test a few cold pressures
Temperatures	TSTC / TSTI/O	surface/carcass temps for the thermal story

The two standard test types you will use: a cornering (pure-slip lateral) test that sweeps slip angle at several loads, cambers and pressures — this is where μ_{peak} , cornering stiffness and load sensitivity come from — and a drive/brake test that sweeps slip ratio. There is also a combined test (slip angle and slip ratio together) for the friction ellipse.

The tools that read it

Table XI-2. Tooling landscape. You can go end-to-end for free.

Tool	Cost	What it does	Fit for Blitz
MFeval	Free / open-source	MATLAB/Simulink evaluator for MF 5.2/6.1/6.2; read a .TIR file, get forces. File Exchange #63618.	Ideal — pairs with your own tool
OptimumTire (OptimumG)	Free to students (historically)	Loads TTC files, fits the Magic Formula with a GUI, exports .TIR	Great for fitting and cross-checking
Your MATLAB tool	Yours	Bins TTC data, fits/scales, extracts μ , $C\alpha$, load-sensitivity, makes your Figs 1–2	The one you must be able to defend
MF-Tyre / MF-Swift	Paid	Industry MF implementation	Not needed for FSAE

The correlation loop that makes all of this real

Lab data $\rightarrow$ Magic-Formula fit $\rightarrow$ scale with $\lambda\mu$ $\rightarrow$ predict grip $\rightarrow$ run a skidpad $\rightarrow$ compare predicted vs measured lateral g $\rightarrow$ adjust $\lambda\mu$. Closing that loop even once, with your own skidpad number, turns “we guessed 0.65” into “we measured our scaling factor.” That is the difference between a mid-pack and a top-tier tyre programme.

Part XII — Processing TTC data and what your analysis tool is doing

This part connects everything to your MATLAB analysis tool. When a judge asks “walk me through your tyre analysis,” this is the answer, step by step. Each step has a reason, and you should be able to give it.

The pipeline

1. **Load the raw data.** TTC rounds come as MATLAB .mat files (and raw text). You load the run(s) for your candidate size and compound. Each file is one test matrix.
2. **Clean and standardise.** Convert to a consistent axis system (SAE or ISO) and units, and discard the warm-up laps and transitions. The machine sweeps slip continuously, so you keep only quasi-steady points. This is where sign-convention bugs hide — verify one known case.
3. **Bin by condition.** Group the data into bins of near-constant vertical load, camber and pressure (e.g. all points at ~660 N, 0° camber, 69 kPa). Within a bin, F_y vs α is a clean curve you can fit. Binning is what turns a cloud of points into the comparison curves.
4. **Fit per bin.** For each bin, fit the Magic Formula (or at least extract slope-at-origin and peak). From the fit you read off cornering stiffness $C\alpha = B \cdot C \cdot D$ and peak $\mu = D/F_z$.
5. **Apply the scaling factor.** Multiply peak grip by $\lambda\mu \approx 0.65$ (friction scaling) so lab numbers become track-representative. Keep raw and scaled clearly separated — mixing them is the most common chart error to avoid.
6. **Extract the comparison metrics.** Plot peak μ vs load (that downward slope = load sensitivity), cornering stiffness vs load, and the $\Delta\mu$ -per-kN bar. Compare candidate sizes on the same axes. These are the comparison-chart panels.
7. **Quality-check.** Check the fit against the raw points, check peak slip angle is physical (a few degrees), check μ is in the believable 1.3–1.8 band after scaling, check both curves use the SAME scaling. Overplot the fitted line on the raw data so the fit is visible.

What each output metric tells you

Table XII-1. The tool’s outputs mapped to the decisions they justify. This table is a cheat-sheet for the design event.

Metric from the tool	Symbol	Design decision it feeds
Peak lateral μ (scaled)	μ_y	Grip ceiling → cornering g → lap time; tyre choice
Cornering stiffness	$C\alpha$	Understeer/oversteer balance; steering response
Load sensitivity ($\Delta\mu/\Delta F_z$)	$\partial\mu/\partial F_z$	How much load transfer hurts → track width, CoG, stiffness split
Peak slip angle	α_{peak}	Slip band the steering ratio and geometry must keep the tyre in
Optimum pressure	P^*	Baseline cold/hot pressures
Camber sensitivity	$C\gamma$	Static camber target

Defending the proxy honestly

When your actual tyre has no test data, a size-matched tyre (say a Hoosier 18×6-10) stands in for it. The defensible statement is narrow and true: “TTC data for this 18” family lets us pick the diameter

and estimate behaviour; our tyre is wider, so the numbers are an estimate to be corrected once we test it.” Judges respect a clearly-stated limitation far more than an over-claimed match — so always say the proxy will be refreshed once the real tyre is tested.

Mini-exercises (do these with your data open)

- **Exercise 1** — pick one load bin and confirm $B \cdot C \cdot D$ from your fit equals the slope you measure by hand over the first 1° . If they disagree, your fit or your units are off.
- **Exercise 2** — read peak μ at 220 N and at 1100 N; the difference over ~ 0.9 kN is your load-sensitivity number. Check it matches the load-sensitivity bar.
- **Exercise 3** — at your true operating load (front ≈ 600 N, rear ≈ 730 N for 270 kg, 45/55), read peak μ there — not at the lightest load. That is the number that belongs in the car-level grip estimate.

Part XIII — From a tyre curve to a lap time: using the data at vehicle level

A tyre curve is only worth points when it becomes a car behaviour. Here are the four bridges from F_y -vs- α to the whole car — the ones a lumped-parameter vehicle model is built on.

1. Grip ceiling → cornering g

At steady-state cornering, lateral acceleration in g is approximately the lateral μ the four tyres can deliver (plus the extra from aero download pressing them harder):

$$a_y (g) \approx \mu_y \times (1 + F_{\text{aero}} / W)$$

where μ_y = scaled peak lateral friction; F_{aero} = downforce at that speed; W = car weight. With no aero, $a_y \approx \mu_y$. This is why your “ ≈ 1.6 g mechanical” is just your scaled μ_y , and why reaching 1.8 g needs real downforce.

2. Cornering stiffness front vs rear → understeer gradient (balance)

Whether the car pushes (understeer) or rotates (oversteer) comes from which axle runs out of grip first, which is set by how much slip angle each axle needs — i.e. the cornering stiffnesses and the loads. The compact measure is the understeer gradient, K .

$$K \propto (W_f / C_{\alpha,f}) - (W_r / C_{\alpha,r})$$

where W_f , W_r = load on front/rear axle; $C_{\alpha,f}$, $C_{\alpha,r}$ = front/rear cornering stiffness (both tyres). $K > 0$ understeer, $K < 0$ oversteer, $K = 0$ neutral. You tune the C-terms with tyre choice and the W-split with roll stiffness.

This is the equation behind a “roll-stiffness distribution designed to induce oversteer and counteract an inherent understeer characteristic.” Moving it changes how load transfer splits front to rear — which changes effective axle grip via load sensitivity. Everything connects.

3. Load transfer → how much the tyres get penalised

In a corner, lateral load transfer moves vertical load to the outside tyres. The total lateral load transfer scales as:

$$\Delta F_z \propto (m \cdot a_y \cdot h_{cg}) / t$$

where m = mass; a_y = lateral acceleration; h_{cg} = centre-of-gravity height; t = track width. Lower CoG and wider track = less transfer = (via load sensitivity) more total grip. How it splits front/rear is set by roll-stiffness distribution.

This is the payoff of [Part VIII](#) made quantitative: your levers against load sensitivity are h_{cg} (low CoG), t (wide track) and m (light car) for total grip, and the roll-stiffness split for balance. When a report says load sensitivity is “managed by ... track width limiting outer-tyre load transfer,” this formula is the mechanism.

4. The yaw-moment (Milliken Moment Method / MMM) diagram — the pro tool

The Milliken Moment Method, MMM, maps the car’s balance across the whole range of lateral grip. You compute the yaw moment (the car’s tendency to rotate) and lateral force for every combination of body slip angle and steer angle, using the tyre curves, and plot them. The result shows your max lateral grip, your stability, your control authority, and your balance at the limit — all on one chart. It is the single most impressive thing a student tyre/VD engineer can bring to a design event, and it is built entirely from the tyre data you now understand. (Build it: Milliken RCVD Ch. 8; the g-g envelope is RCVD Ch. 9.)

Part XIV — Tyre selection, done the way judges respect

“Why this tyre?” is the first question you will get. A strong answer is a documented decision matrix driven by the data from [Part XII](#), not a preference. Here is the structure.

The decision variables

- **Diameter / rim** — smaller (10") lowers unsprung mass, rotational inertia and the car; many teams move 13"→10" for exactly this and quantify it (settling time, contact-patch variation). Bigger carries more load with less pressure sensitivity.
- **Width** — wider usually means higher peak μ but more mass, inertia, drag and sometimes load sensitivity. A 7.7" tyre vs a 6.0" Hoosier proxy is this trade.
- **Compound** — softer = higher peak grip and faster warm-up (good for short FSAE events) but faster degradation over endurance. LC0, R20 etc. are points on this curve.
- **Data availability** — a tyre you have TTC data for is worth more than a marginally better tyre you must guess at. Data availability is a legitimate, defensible selection criterion.
- **Cost / availability** — cost and supply are real constraints and judges accept them if you are honest. A “budget tyre now, premium tyre when funded” position is exactly this — state it plainly.

The method

1. **Shortlist.** List candidate tyres (size × compound) you can actually buy.
2. **Score on data.** For each, pull peak μ (scaled), cornering stiffness, load sensitivity and peak slip angle at YOUR operating loads from the TTC/your tool.
3. **Add the practical rows.** Add weighted rows for mass/inertia, warm-up, durability, cost and data availability.
4. **Weight by what wins points.** Weight the rows by event points (skidpad/autocross are cornering-heavy → weight peak μ and warm-up highly).
5. **Decide and document.** The matrix picks the tyre; you then quote the two or three numbers that decided it. That is a design-event answer.

Your actual situation, framed well

When you lack data for your actual tyre, you score a size-matched proxy (say the 18" Hoosier family) to justify the DIAMETER, choose your tyre in that size on cost, and flag the width gap and the plan to test. That IS a defensible method — just add a number for the width penalty and a line on when you test.

Part XV — Setup and testing: pressures, pyrometry, skidpad correlation

Analysis sets your starting point; testing earns the last tenths and the data that makes next year's analysis real. A simple, disciplined test programme beats a fancy one you cannot execute.

A minimal, high-value FSAE tyre test programme

1. **Pressure & camber sweep (half a day)** — with a tyre pyrometer, measure cold pressure, run 3–5 hard laps, come in and immediately read inner/middle/outer tread temps and hot pressure. Adjust pressure and camber per Table IX-1. Repeat. This alone finds most of your lap time.
2. **Skidpad for grip correlation** — run the constant-radius skidpad at a range of speeds; the steady lateral g you achieve is your real μ_y . Compare to your tool's prediction and back out your true scaling factor $\lambda\mu$. Now 0.65 is measured, not assumed.
3. **Step-steer for response** — a sharp steering input at constant speed; how long the car takes to settle is the transient-response / relaxation story (often reported as a “step steer ... settling time”). Lower inertia and the right tyre sharpen it.
4. **G-G from logged laps** — log lateral and longitudinal g and build the g-g plot ([Part VII](#)). Where the cloud does not reach the edge is grip the driver is leaving unused — coaching feedback and setup feedback at once.

What to record every single run

Cold pressure, hot pressure, tread temps (I/M/O, each corner), ambient and track temp, camber, toe, time since the tyre was last run, number of heat cycles on the tyre, and the driver. Tyres age by heat cycles (the rubber hardens), so an “old” set that has done many cycles will not match your fresh-tyre data — track this or you will chase ghosts.

The habit that compounds

Every run logged in a consistent sheet becomes next year's model input. The reason top teams look like they “just know” their tyres is years of this logbook. Start it now, even crude; it is the highest-return, lowest-cost thing the tyre lead can institute, and it is exactly the kind of process the design event scores.

Part XVI — Reading a vehicle-dynamics report: common terms decoded

Strong FS suspension & tyre write-ups are dense with jargon. Here is a passage of typical phrasing translated into what you now know — read the phrase, then the decode. By the end none of it should be opaque.

Table XVI-1. Common FS tyre/VD phrasing, decoded. Every term maps to a Part above.

Typical phrasing you'll see	What it actually means
"Hoosier 6/18-10 LC0 tyres"	A Hoosier slick, 6" wide, 18" OD, 10" rim, LC0 compound. Same size nomenclature as an 18×6-10 (Part II).
"low unsprung mass and yaw inertia"	Unsprung mass = everything not held up by the springs (wheel, tyre, upright, brake). Yaw inertia = resistance to the car rotating about its vertical axis. Small 10" tyres cut both → wheels follow bumps better and the car changes direction more eagerly.
"corner mass ... 1.97 kg ... reduced RMS contact-patch force variation by 9%"	Corner mass = mass at one wheel station. Lighter unsprung mass lets the tyre stay in better contact over bumps, so the vertical load wobbles less (lower root-mean-square variation) → steadier grip (Part I: force needs steady load).
"step steer ... settling time 11% faster for 10" vs 13"	Step-steer = snap the wheel and watch how fast the car settles (Part XV). Lower inertia + relaxation length (Part V) → quicker, crisper response.
"LC0 ... faster tyre temperature build"	The compound reaches its grip temperature window sooner (Part IX) — worth real points on a cold FSAE skidpad.
"99% of peak lateral force over an 8° slip-angle range ... driveable ... amateur drivers"	A broad, forgiving force peak (Part V): the driver can be several degrees off the ideal slip angle and still get ~full grip. Great for student drivers.
"lumped parameter model"	A simplified whole-car model (masses, stiffnesses, tyre C_α) used to predict handling fast — fed by exactly the tyre numbers from your analysis (Part XIII).
"weight transfer, wing size, driving line" (choosing wheelbase/track)	Wider track lowers load transfer (Part XIII) but costs aero/package and changes the line through tight corners — a compromise.
"weight distribution 49/51, CoG height 270 mm"	Front/rear load split and how high the mass centre sits. CoG height drives load transfer directly (Part XIII). A typical FSAE car is ~45/55 and ~270 kg.
"dynamic camber change"	How much camber the wheel gains/loses as the suspension moves and the body rolls (Part IX) — you design static camber around it.
"motion ratio"	How much the spring/damper moves per unit wheel movement — sets effective wheel rate and ride frequency; not a tyre term but it changes load transfer behaviour.
"roll stiffness distribution 30–73% front ... induce oversteer"	How roll resistance (springs + anti-roll bars) splits front/rear. Moving it changes which axle transfers more load → shifts balance via load sensitivity (Part VIII, XIII).
"removing front ARB worsens contact-patch variation under roll"	Anti-roll bar (ARB) = a spring that resists body roll only. Without the front bar the front tyres' load fluctuates more in roll → less steady grip.
"4-post rig testing"	A lab rig that shakes all four wheels to measure how well the tyres stay loaded over bumps — validating the contact-patch-variation claims.

Part XVII — Sounding like an expert in the design event

The judges are mostly checking whether you understand your own numbers and their limits. A few habits make you read as an expert:

- **Cite the source of every number** — every tyre claim should trace to TTC data, your fit, or your own track test — and you should say which. “We measured / we fitted / we assumed, and here is why” beats any confident round number.
- **State limitations before they ask** — volunteering “our tyre is wider than the proxy, so these are estimates until we test” earns more trust than a seamless story. Judges hunt for over-claims; pre-empt them.
- **Use the right number, not the best-looking one** — quote peak μ at your operating load, not the head-line light-load value; quote cornering stiffness and load sensitivity, not just “grip.” Specificity signals understanding.
- **Speak the vocabulary** — “friction scaling factor $\lambda\mu$,” “cornering stiffness $C\alpha$,” “load sensitivity,” “pneumatic trail,” “peak slip angle,” “relaxation length.” Each is defined in this guide; using them correctly is the fastest credibility.
- **Close the loop to lap time** — tie tyre numbers to the car: “this load sensitivity is why we set track width and roll distribution thus,” “this peak slip angle is why our steering ratio is 6:1.” That connection is what separates a tyre technician from a vehicle-dynamics engineer.

If you only remember five phrases

(1) Grip is a surface, not a number. (2) Force comes from slip. (3) μ falls as load rises — load sensitivity. (4) One grip budget, shared — the friction ellipse. (5) Lab grip $\times$ scaling factor = track grip. Everything in this guide hangs off these five.

BOOK TWO

| FORMULA STUDENT VEHICLE DYNAMICS

Vehicle Dynamics & Suspension

Mass, load transfer and balance; suspension and steering geometry; springs and dampers; how every decision couples; and telemetry.

Part A — Vehicle dynamics fundamentals

Vehicle dynamics is the study of how forces make the car accelerate, brake, corner and ride. All four come from the tyres ([Book One](#)); this book is about everything that decides how much force each tyre gets to make and when.

Sprung and unsprung mass

- **Sprung mass** — everything held up by the springs — chassis, engine, driver, fuel, bodywork, aero. It is what rolls, pitches and heaves on the suspension.
- **Unsprung mass** — everything between the road and the springs — wheel, tyre, upright, hub, brake disc, and roughly half the arms/pushrod/driveshaft. It is what has to follow the road surface over bumps.

Why the split is the first thing a VD engineer internalises: the sprung mass is what you control the motion of (with springs and dampers), and the unsprung mass is what degrades grip when it is heavy — a heavy wheel cannot follow a bump fast enough, so the tyre load fluctuates and average grip drops. This is the “contact-patch force variation” that strong FS teams obsess over, and the whole reason they chase low unsprung mass with 10” wheels and lightweight uprights.

The vehicle axis system and the six motions

Fix a right-handed axis set at the car’s centre of gravity: x forward, y to the right, z down (the same SAE convention as the tyre, [Book One Part III](#)). The car has six degrees of freedom — three translations and three rotations:

Table A-1. The six rigid-body degrees of freedom. Roll, pitch and yaw are the three you tune.

Motion	Axis	Name	Caused mainly by
Translation	x	Surge (accel/brake)	Traction, braking, drag
Translation	y	Sway (lateral)	Cornering force
Translation	z	Heave (vertical)	Bumps, aero load, weight
Rotation	x	Roll	Lateral load transfer in a corner
Rotation	y	Pitch	Braking (dive) and acceleration (squat)
Rotation	z	Yaw	The car rotating to point into a corner

Roll, pitch and yaw are where suspension design lives. Roll and pitch you resist and damp (springs, bars, dampers, geometry); yaw you shape (that is balance — [Part D](#)). Heave you support (ride springs). Everything in Parts E–G is about controlling these motions so the tyres stay in their happy window as much of the lap as possible.

Part B — Mass properties and how to measure them

You cannot predict load transfer, balance or inertia response without three numbers: total mass, where the centre of gravity (CoG) is, and the car's moments of inertia. Guessing these is the single most common reason a lap-sim does not match the real car. Measure them.

Weight distribution and CoG position

The CoG has three coordinates, and each matters for a different reason:

- **Longitudinal (x)** — front/rear split (your 45/55). Sets static axle loads, hence how much grip each axle starts with and how braking/traction load-transfer plays out.
- **Lateral (y)** — left/right split. Should be near 50/50; asymmetry (driver offset, battery) is corrected with corner weighting ([Part M](#)).
- **Vertical (z)** — CoG height, h_{cg} . The single most important number for load transfer: it multiplies every transfer term. Lower is almost always better. FSAE cars are typically ~250–320 mm; e.g. ~270 mm is in range.

Measuring CoG height (the tilt method — do this)

Put the car on four scales level, record axle loads. Then raise one axle by a known height so the car tilts by angle θ , and re-weigh. The load that shifts between axles tells you the CoG height:

$$h_{cg} = (\Delta W_f \cdot L) / (W \cdot \tan\theta) + r_{loaded}$$

where ΔW_f = increase in front-axle weight when the REAR is raised; L = wheelbase; W = total weight; θ = tilt angle; r_{loaded} = loaded wheel radius (adds the axle-to-ground height). Bigger tilt = better accuracy.

CoG-height measurement — the traps

Lock out the suspension (solid links or zip-tied dampers) and fully inflate tyres, or the car squats as you tilt and corrupts the result. Strap the driver (or a dummy of their mass) in the seat — the driver is ~25% of the car and moves the CoG a lot. Remove fuel slosh (run a known fixed amount). Tilt as far as you safely can (15–25°); a 5° tilt gives a noisy answer. Repeat three times and average.

Moments of inertia — why the car feels eager or lazy

Mass tells you how hard the car is to speed up; moment of inertia tells you how hard it is to rotate. The one that dominates handling feel is yaw inertia, I_{zz} — resistance to rotating about the vertical axis.

- **Yaw inertia (I_{zz})** — a low yaw inertia car changes direction eagerly (good for a tight autocross), responds faster to steering, and settles quicker after a flick. This is why teams pull mass toward the centre (engine, driver, fuel central) and off the ends (10" vs 13" wheels cut mass far from the centre).
- **Roll (I_{xx}) and pitch (I_{yy}) inertia** — resistance to roll and to pitch; affect how quickly the car takes a set in roll and dives/squats. Less critical to tune than yaw but part of the transient story.

Measure yaw inertia with a bifilar (two-wire) or trifilar pendulum: hang the car (or measure it on a rotating platform), set it oscillating about the vertical axis, and time the period — the period squared gives the inertia. A quoted “moment of inertia measurement” is exactly this. If you cannot measure it, estimate from CAD mass properties and flag it as an estimate.

The radius-of-gyration intuition

Two cars of equal mass and wheelbase can feel completely different because one carries its mass at the ends (big yaw inertia, lazy) and one at the centre (small yaw inertia, darty). The “dumbbell vs ball” picture is the whole reason mass centralisation is a design goal, not just mass reduction. A 2 kg wheel saved at each corner helps twice: less unsprung mass AND less yaw inertia.

Part C — Load transfer: the master concept

Load transfer is the redistribution of vertical tyre load when the car accelerates, brakes or corners. Grip depends on vertical load ([Book One](#)), so load transfer is how the car decides, instant to instant, which tyres are strong and which are weak. Master this and most of suspension design becomes obvious.

Longitudinal load transfer (braking and acceleration)

Brake and the nose dives — load moves forward onto the front tyres; accelerate and it moves rearward. The amount is purely a mass property, independent of springs:

$$\Delta F_{z_long} = (m \cdot a_x \cdot h_{cg}) / L$$

where m = mass; a_x = longitudinal acceleration; h_{cg} = CoG height; L = wheelbase. Note: springs set HOW it arrives (dive/squat) and WHEN, but not HOW MUCH.

Consequences you design for: the front brakes do more work (so bigger front discs/pads and a forward brake bias), the rear can light up under acceleration (traction limited — [Book One Part VI](#) and the LSD), and a long wheelbase or low CoG reduces dive/squat. This is also why anti-dive/anti-squat geometry exists ([Part E](#)).

Lateral load transfer (cornering) — and its three paths

In a corner, load moves from the inside tyres to the outside tyres. The TOTAL amount, summed over both axles, is again a mass property:

$$\Delta F_{z_lat} \text{ (total)} = (m \cdot a_y \cdot h_{cg}) / t$$

where a_y = lateral acceleration; t = track width. Lower CoG, wider track, less mass → less total transfer → (via load sensitivity, [Book One Part VIII](#)) MORE total grip.

But — and this is the key that unlocks balance tuning — that total transfer reaches the tyres by three different paths, and you control how it splits front-to-rear by changing them:

Table C-1. The three paths of lateral load transfer at one axle. The elastic path is how you tune balance; the geometric path is instant and sets initial response.

Path	How the load gets there	Speed	Your tuning lever
Elastic (sprung)	Through the springs and anti-roll bars as the body rolls	Builds as the car rolls (slight lag)	Spring rates and ARB stiffness — the MAIN balance knob
Geometric (sprung)	Through the suspension links, reacted at the roll centre — no body roll needed	Instant	Roll-centre heights (geometry)
Unsprung	Direct from the unsprung mass to the tyre	Instant	Unsprung mass and its CoG height (small)

The idea that makes balance tunable at all

Total lateral load transfer is fixed by mass, CoG height, track and how hard you corner — you cannot change it with springs. What you CAN change is how much of it goes through the FRONT axle versus the REAR, by making one end roll-stiffer than the other. Because of tyre load sensitivity, the axle that takes more load transfer loses relatively more grip → that end slides first. Stiffen the front → more understeer; stiffen the rear → more oversteer. That is the master balance knob, and it has a name: Total Lateral Load Transfer Distribution (TLLTD).

Part D — Steady-state balance: understeer, oversteer, and the knob

“Balance” is which end of the car runs out of grip first. It is the thing drivers feel, the thing that wins autocross, and the thing you spend the whole setup chasing. Here is the complete picture.

Understeer, oversteer, neutral

- **Understeer** — the front tyres reach their limit first; the car runs wide; the driver must lift or add steering that does nothing. Safe, stable, but slow if excessive. Caused by too much front load transfer, too-soft front tyres/springs relative to rear, too little front grip.
- **Oversteer** — the rear tyres reach their limit first; the back steps out; the car rotates (or spins). Fast in the right hands, unstable if excessive. Caused by too much rear load transfer, rear-soft balance, or power-on at a rear-heavy car.
- **Neutral** — both axles reach the limit together — the theoretical ideal for maximum grip, usually tuned to a hair of stable understeer for driver confidence.

The understeer gradient, quantified

The formal measure ([Book One Part XIII](#) introduced it) compares how much slip angle each axle needs:

$$K \propto (W_f / C_{\alpha,f}) - (W_r / C_{\alpha,r})$$

where W_f , W_r = axle loads; $C_{\alpha,f}$, $C_{\alpha,r}$ = front/rear axle cornering stiffness. $K > 0$ understeer, $K < 0$ oversteer, $K = 0$ neutral. Units commonly deg/g.

You can measure K directly on the car ([Part L](#)): drive a constant-radius circle at increasing speed and plot steering angle against lateral g . A rising line = understeer, flat = neutral, falling = oversteer. This is the single most important handling measurement you can make, and it closes the loop between your sim and reality.

How balance moves around — the four effects you must expect

- **With speed** — as speed rises, aero adds load (if you have downforce) and the understeer gradient shifts; a car balanced at 40 km/h may push at 90. Your aero CoP-vs-CoG discussion is exactly this.
- **With roll-stiffness split (TLLTD)** — stiffen the front relative to the rear (springs or ARB) and the front takes more load transfer → more understeer. This is the knob you turn most. A quoted “30–73% front roll-stiffness range” is the adjustment range of exactly this.
- **With the differential** — a locked/limited-slip diff adds understeer on entry (both rears driven) and can add power-on oversteer on exit. Your Drexler LSD is a balance tool, not just a traction tool.
- **With weight distribution** — on a rear-heavy car (yours, 45/55) the rear axle is more loaded and more load-sensitive, so it needs proportionally LESS roll stiffness to avoid over-transfer — hence “rear roll centre higher than front” and careful rear bar sizing.

The balance mantra

Grip you set with magnitude of load transfer (low CoG, wide track, light car). Balance you set with distribution of load transfer (front-vs-rear roll stiffness, roll-centre heights, diff, tyre sizes). When the car is slow AND balanced, chase magnitude. When it is gripping but pushing or loose, chase distribution. Never confuse the two — teams waste whole test days turning the wrong knob.

Part E — Suspension kinematics: the geometry, parameter by parameter

Kinematics is the geometry of how the wheel moves relative to the chassis — before any spring force enters. It decides the camber the tyre sees, where load transfer is reacted, how the steering feels, and how the car dives and squats. Get the geometry wrong and no spring or damper setting will save you. This is the longest part of the book because it is where the most permanent decisions are made: springs you can change at the track, hardpoints you cannot.

Why FSAE runs double wishbones

A double-wishbone (double A-arm) suspension uses two control arms per wheel, giving the designer near-total control of the camber curve and roll-centre location through the choice of arm lengths and angles. That control is why essentially every competitive FSAE car uses it, usually actuated by a pushrod or pullrod onto an inboard spring/damper to cut unsprung mass and clean up aero (most competitive FSAE cars do this).

- **Pushrod** — rod in compression, spring/damper mounted low/inboard; simple, common, slightly higher CoG for the spring mass. Your choice.
- **Pullrod** — rod in tension, spring/damper mounted high; lowers the inboard mass, trickier packaging.
- **Strut** — fewer parts, lighter, but couples spring and steering and gives poor camber control — rare in FSAE for good reason.

The geometric parameters — each one: what it is, what it does, typical values

Camber, camber gain and the camber curve

Static camber is the wheel's lean at rest ([Book One Part IX](#)). Camber gain is how camber changes as the wheel moves up (bump) — the slope of the camber curve. You want the outside wheel to stay near its ideal (slightly negative) camber as the body rolls, which means the geometry must ADD negative camber in bump to fight the positive camber that body roll would otherwise give the outer wheel.

(camber the loaded outer tyre sees) = static camber + bump camber gain − body roll angle

where Design target: the loaded outer tyre ends near 0° to −1° dynamic camber at max roll. Pyrometry ([Book One Part IX](#)) confirms it.

Camber gain comes from short, angled upper arms (shorter front-view swing arm = more gain). But more camber gain costs roll-centre stability — see the trade-off box below. Typical FSAE static camber: ~−1° to −3° front; camber gain tuned so dynamic camber lands right.

Instant centre, front-view swing arm, and the roll centre

Project the two control arms in the front view; where their lines cross is the instant centre (IC) — the point the wheel instantaneously pivots about. The distance from the tyre contact patch to the IC is the front-view swing-arm (FVSA) length; it sets camber gain (short FVSA = lots of gain). Draw a line from each tyre's contact patch to its IC; where those two lines cross on the car's centreline is the roll centre (RC) — the point about which the sprung mass rolls at that axle.

- **Roll-centre height** — the height of the roll centre sets how much lateral load transfer goes through the GEOMETRIC (instant) path vs the elastic (spring) path, and sets jacking (below). Low RC = more roll, more elastic transfer, more tunable, but more camber/aero movement. High RC = less roll, quick geometric response, but jacking and harshness. FSAE RCs are usually just above or below ground, within ~0–100 mm.

- **Roll axis** — the line joining the front and rear roll centres. The sprung mass rolls about this axis; the perpendicular distance from the sprung CoG to the roll axis is the roll moment arm that drives roll angle ([Part F](#)). Running the “rear roll centre higher than the front” tilts this axis to bias balance.

Jacking — the high-roll-centre trap

A roll centre well above ground makes the lateral force at the contact patch push the sprung mass UP (the car jacks itself up mid-corner). That raises CoG, reduces grip, and feels nervous — worst with a very high RC and a swing-axle-like geometry. Keep roll centres low and their migration small. This is a classic FSAE mistake: chasing “less roll” with a high RC and getting a car that climbs onto its outside tyre and snaps.

The central kinematic trade-off

Short angled arms give lots of CAMBER GAIN (good: keeps the tyre flat in roll) but move the INSTANT CENTRE a long way as the wheel travels, so the ROLL CENTRE MIGRATES up/down and sideways (bad: balance and jacking change mid-corner). Long arms give a STABLE roll centre but little camber gain (tyre leans over in roll). You cannot maximise both. The usual student-car call is a stable roll centre with moderate bump camber, because a migrating roll centre changes the balance unpredictably mid-corner, which is harder to drive than slightly imperfect camber. Be ready to defend that trade in the design event.

Caster and mechanical trail

Caster is the forward/backward tilt of the steering axis viewed from the side. Positive caster places the steering-axis’s ground intersection ahead of the contact patch, creating mechanical trail — a lever arm that makes the aligning moment ([Book One Part V](#)) pull the wheels straight. Caster also adds negative camber to the outside wheel as you steer (free camber in corners) and is the main source of steering self-centring and “feel.”

- **More caster** — more straight-line stability, more self-centring, more dynamic camber in corners — but heavier steering. FSAE typically $\sim 3\text{--}7^\circ$.
- **Trail** — mechanical trail adds to pneumatic trail in the steering torque the driver feels; set it so the wheel still has feel near the limit (where pneumatic trail collapses to zero — [Book One Part V](#)).

Kingpin inclination (KPI/SAI) and scrub radius

Kingpin inclination (KPI, also steering-axis inclination SAI) is the inward lean of the steering axis viewed from the front. It provides self-centring (the car rises slightly when steered, so gravity straightens it) and, with the wheel offset, sets the scrub radius — the distance from where the steering axis hits the ground to the centre of the contact patch.

- **Zero/small scrub** — tends toward zero steering torque from braking/traction forces (good), but gives no feedback from one-wheel bumps; small positive scrub is usually chosen for feel and packaging.
- **Large scrub** — makes braking/traction force differences (e.g. one wheel on a bump, or brake imbalance) kick back through the wheel. Keep scrub radius small in magnitude.

KPI fights camber: a lot of KPI puts positive camber on the outside wheel as you steer, undoing caster’s help. Use the least KPI that packages the upright/bearing/brake, and lean on caster for the dynamic camber instead.

Toe and bump steer

Toe is whether the wheels point slightly in (toe-in) or out (toe-out) viewed from above. Bump steer is unwanted toe change as the wheel moves in bump/roll — caused by the steering tie-rod not matching the suspension arc.

- **Toe** — front toe-out sharpens turn-in (outside wheel leads the car in) but adds straight-line nervousness and tyre scrub/drag; front toe-in adds stability. Rear toe-in adds stability (the usual choice); rear toe-out is dangerous (snap oversteer).

- **Bump steer** — should be designed to NEAR ZERO. Any bump steer means the car steers itself over bumps and in roll — random, un-tunable, and a confidence-killer. Getting the tie-rod inner/outer points on the right arc is a non-negotiable kinematic task. Strong teams explicitly “minimise bump steer through careful tie-rod placement.”

DO	DON'T
✓ Design bump steer to within $\sim 0.05^\circ$ per mm of travel (effectively zero) and verify it in your kinematics software AND on the car with the steering fixed.	✗ Never set rear toe-out — it causes snap oversteer at the worst moment.
✓ Run a little rear toe-in for stability on a rear-heavy car.	✗ Don't ignore bump steer because “the sim looked fine” — build tolerance means measure it on the real car.
✓ Make toe adjustable with shims/turnbuckles — you WILL tune it.	✗ Don't chase turn-in with huge front toe-out — it scrubs grip and overheats the fronts.
✓ Check toe under load (it changes with compliance), not just static.	✗ Don't forget toe eats tyre temperature — cross-check with pyrometry.

Ackermann steering geometry

In a corner the inside wheel turns a tighter radius than the outside, so it should steer to a larger angle. Ackermann geometry is how much the steering linkage builds in that difference. 100% (“full”) Ackermann exactly matches the low-speed geometric ideal; 0% (parallel steer) turns both wheels equally; some cars run anti-Ackermann (outside turns more) to suit the tyre’s slip-angle needs at high load.

Why it is not obvious: tyres want slip angle, and the heavily-loaded outside tyre (doing most of the work) may want a DIFFERENT slip angle than the geometric ideal. FSAE autocross is low-speed and tight, so most teams run partial-to-high Ackermann to reduce inside-tyre scrub — a quoted “49% Ackermann” chosen for “low-speed agility” is this compromise. Tie the choice to your tyre’s peak-slip-vs-load data ([Book One](#)), not to a textbook default.

Anti-dive, anti-squat, anti-lift (longitudinal geometry)

By tilting the side-view control-arm lines, you can make the suspension react some of the braking/acceleration load transfer through the links instead of the springs — reducing dive under braking (anti-dive), squat under acceleration (anti-squat), and lift. It is expressed as a percentage of the load transfer taken geometrically.

- **Anti-dive** — useful to keep the aero platform and brake balance stable, but too much makes the suspension harsh under braking (it stops absorbing bumps while braking) and can cause the car to “push” over bumps. FSAE typically runs modest anti-dive ($\sim 10\text{--}30\%$).
- **Anti-squat** — keeps the rear from squatting on power; too much causes harshness and wheel-hop on bumpy exits. Usually kept low on FSAE cars.

Rule of thumb: start near zero-to-moderate anti geometry; it is a refinement, not a foundation. Prioritise the lateral geometry (camber, roll centre, bump steer) first.

Motion ratio (installation ratio) — the bridge to springs

The motion ratio (MR), or installation ratio, is how much the spring/damper moves for a given wheel movement. It links wheel behaviour to spring behaviour and it is squared when converting rates, so small errors matter a lot.

$$\text{MR} = (\text{spring/damper travel}) / (\text{wheel travel}) \quad K_{\text{wheel}} = K_{\text{spring}} \times \text{MR}^2$$

where MR usually < 1 for a pushrod car (spring moves less than the wheel); K_{wheel} = rate felt at the wheel; K_{spring} = coil rate. Because of the square, an MR of 0.6 turns a 60 N/mm spring into only ~ 21.6 N/mm at the wheel.

- **Design the rocker so** — aim for a near-constant MR through travel (a “linear” rocker), or deliberately rising-rate if you want more support at high load. A wildly varying MR makes the wheel rate and damping change through travel in ways that are hard to predict and tune.

- **Verify** — measure the real MR on the car (pushrod/damper pot vs wheel displacement) — the built MR is never exactly the CAD MR, and everything downstream (wheel rate, ride frequency, damping) depends on it.

Kinematics summary — the cheat sheet

Table E-1. Suspension kinematics at a glance. Hardpoints are permanent — get these right before you cut tube.

Parameter	Primary effect	Typical FSAE	Watch out for
Static camber	Dynamic camber under roll	-1° to -3° F	Inner-edge overheating if too much
Camber gain	Keeps tyre flat in roll	Tuned to $\sim 0^\circ$ dyn	Trades against RC stability
Roll-centre height	Geometric vs elastic transfer; jacking	~ 0 –100 mm	Jacking if too high
RC migration	Mid-corner balance stability	Keep small	Short arms migrate a lot
Caster	Self-centring, feel, corner camber	3 – 7°	Heavy steering if excessive
KPI/SAI	Self-centring, scrub radius	As low as packaging allows	Adds positive camber on steer
Scrub radius	Kickback, feedback	Small magnitude	Harsh over one-wheel bumps if large
Toe (front/rear)	Turn-in vs stability	~ 0 ; rear slight toe-in	Rear toe-out = snap oversteer
Bump steer	Self-steer over bumps	≈ 0	Must verify on car
Ackermann	Inside/outside steer match	partial–high	Tie to tyre slip data
Anti-dive/squat	Dive/squat, platform	0–30%	Harshness if too much
Motion ratio	Wheel rate & damping	~ 0.5 –1.0	Squared — verify on car

Part F — Springs, roll stiffness and ride

Kinematics decided how the wheel moves; now the springs decide how hard it resists moving. Springs do two jobs at once — support the car over bumps (ride) and resist body roll (roll stiffness) — and the art is setting both without an anti-roll bar fighting you.

From spring rate to wheel rate to ride frequency

What the tyre feels is the wheel rate, not the coil rate — converted through the motion ratio squared ([Part E](#)). The clean way to choose springs is not in N/mm but in ride frequency, the natural bounce frequency of the sprung corner, because it normalises for mass and lets you compare front to rear sensibly.

$$f_{\text{ride}} = (1 / 2\pi) \cdot \sqrt{(K_{\text{wheel}} / m_{\text{sprung, corner}})}$$

where f_{ride} = ride frequency (Hz); K_{wheel} = wheel rate (N/m); $m_{\text{sprung, corner}}$ = sprung mass carried at that corner (kg). Choose f first, then back out the spring.

Typical targets:

- **Ride frequency** — non-aero FSAE cars commonly run ~2.5–3.5 Hz; cars with real downforce go stiffer (3.5–5 Hz) to control ride-height change under aero load. Yours is non-aero-dominated, so the lower band.
- **Front vs rear** — a common choice is the FRONT slightly softer than the rear (e.g. front ~0.1–0.3 Hz lower) so the car takes a “flat ride” over bumps — the rear catches up to the front’s pitch motion. Some race setups invert this; decide with data, not dogma.

Why stiffer is not automatically faster

Beginners stiffen everything to “reduce roll” and “sharpen the car,” then wonder why it has no grip on bumpy autocross. A too-stiff car cannot keep its tyres in contact over bumps — contact-patch load fluctuates, average grip drops ([Part A](#), and the contact-patch-variation theme in both reports). Run the SOFTEST springs that still control ride height, roll and the aero platform. Mechanical grip lives in compliance; you buy roll control back with anti-roll bars, not brute spring rate.

Roll stiffness and the roll gradient

Roll stiffness is how strongly the car resists leaning in a corner, in N·m of roll-resisting torque per degree of roll. It comes from the springs (acting at track width) plus the anti-roll bars. The headline number is the roll gradient — degrees of body roll per g of lateral acceleration:

$$\text{roll gradient} = (W_{\text{sprung}} \cdot H_{\text{arm}}) / K_{\text{roll, total}} \text{ [deg/g]}$$

where W_{sprung} = sprung weight; H_{arm} = perpendicular distance from sprung CoG to the roll axis; $K_{\text{roll, total}}$ = total roll stiffness. Target is usually ~1.0–2.0 deg/g for an FSAE car — enough compliance for grip, little enough to keep camber and aero sane.

Too much roll and the outer tyres go to bad camber and the car feels vague and late; too little roll (super-stiff) and you are back to the no-mechanical-grip problem. Pick a roll-gradient target first, then split the stiffness front-to-rear to set balance (next).

Anti-roll bars and TLLTD — the balance knob, sized

An anti-roll bar (ARB) adds roll stiffness at one end WITHOUT adding ride stiffness — it only resists the two wheels moving in opposite directions (roll), not together (bump). That is exactly what you want: tune roll stiffness distribution for balance while keeping ride soft for grip. Removing an ARB dumps that end’s

roll control back onto the springs and raises contact-patch variation — which is why teams that test it usually keep the front bar.

The balance link: the fraction of TOTAL roll stiffness at the front = the fraction of lateral load transfer taken at the front = Total Lateral Load Transfer Distribution (TLLTD).

$$\text{TLLTD_front} \approx K_{\text{roll, front}} / (K_{\text{roll, front}} + K_{\text{roll, rear}})$$

where More front roll stiffness → higher front TLLTD → front loses relatively more grip (load sensitivity) → more understeer. The adjustable range of your bars (e.g. a ~5% window) is your in-session balance trim.

How to actually set it

Pick total roll gradient for grip and camber control. Then set front-vs-rear roll-stiffness split to the balance you want (start ~just-forward of neutral for stable understeer). Make at least one ARB blade-adjustable so you can trim balance between runs WITHOUT changing total roll or ride. Then verify the real roll gradient on the car from damper-pot data ([Part L](#)) — the built number never equals the CAD number.

DO	DON'T
✓ Choose springs by ride frequency, then convert through the measured motion ratio.	✗ Don't stiffen everything to kill roll — you kill mechanical grip with it.
✓ Run the softest springs that still control ride height, roll and (if present) aero platform.	✗ Don't forget MR is squared — a small rocker error ruins your rate.
✓ Use ARBs — not spring rate — to set front/rear balance.	✗ Don't set balance with asymmetric spring rates front-to-rear if a bar can do it cleaner.
✓ Make one bar adjustable and log roll gradient to verify.	✗ Don't trust CAD sprung-corner masses — corner-weigh the real car.
✓ Keep a baseline setup you can always return to.	✗ Don't change two things at once when tuning — you learn nothing.

Part G — Dampers: what they really control, and how to tune them

Springs set where the car sits and how hard it resists; dampers (shock absorbers) control how fast it gets there and how it settles. They are the least understood and most powerful tuning tool on the car, because they act on VELOCITY, not position — so they shape transients (turn-in, bumps, kerbs) without changing steady-state balance or ride height.

Damping ratio and critical damping

The reference point is critical damping — the damping that returns the mass to rest as fast as possible without overshoot. Real cars run a fraction of it, the damping ratio ζ (zeta).

$$\zeta = c / (2 \cdot \sqrt{(K_{\text{wheel}} \cdot m_{\text{sprung, corner}})})$$

where c = damping coefficient at the wheel; $\zeta < 1$ underdamped (overshoots), $\zeta = 1$ critical, $\zeta > 1$ overdamped (sluggish). Race cars typically run $\zeta \approx 0.6$ – 0.7 for body control, with rebound higher than bump.

The two velocity regimes — the key mental model

A damper behaves differently at low shaft speed than high, and they control different things. The split is usually around 25–50 mm/s of damper velocity.

Table G-1. Low-speed vs high-speed damping. Most “balance” tuning with dampers is in the LOW-speed range; most “grip over bumps” is HIGH-speed.

Regime	Shaft speed	Comes from	Controls	If too much →
Low-speed	< ~25–50 mm/s	Body motions: roll, pitch, heave in corners	Platform control, transient balance, “response”	Harsh, skates over small bumps, nervous
High-speed	> ~50 mm/s	Wheel motions: bumps, kerbs, road texture	Keeping the tyre on the road (contact-patch load)	Wheel can’t follow road → grip loss, crashy

Compression (bump) vs rebound

- **Compression (bump)** — resists the wheel moving UP into the car. Too much and bumps punch through to the chassis and unload the tyre on the far side of the bump.
- **Rebound** — resists the wheel extending back DOWN. Controls how the sprung mass settles after it has been lifted/rolled. Usually set higher than compression. Too much rebound “jacks down” — the car ratchets lower over successive bumps and loses travel and grip.

How dampers tune balance without touching the springs: a stiffer damper at one end during the transient makes that end take its load transfer FASTER, so it reaches its grip limit sooner during turn-in. More front low-speed compression/rebound → entry understeer; more rear → entry oversteer/rotation. This is why dampers are the entry/exit-balance tool while ARBs are the mid-corner (steady-state) tool.

A damper tuning process you can actually run

1. **Baseline the springs/bars** — set springs and ARBs first (steady-state balance and ride). Dampers do not fix a bad spring choice.
2. **Log velocities** — with damper-position sensors ([Part J](#)), log a lap and build a damper-velocity histogram for each corner. A healthy histogram is roughly bell-shaped, centred near zero, not clipped at the ends.
3. **Set low-speed for control** — use low-speed damping to trim transient balance and platform; a common start is rebound a bit above compression, $\zeta \sim 0.6\text{--}0.7$.
4. **Set high-speed for grip** — use high-speed damping (and blow-off/“knee”) so the wheel follows kerbs and bumps without unloading the tyre; watch contact-patch load variation (from pushrod strain, [Part L](#)).
5. **Iterate back-to-back** — change one adjuster a few clicks, drive the same corner, compare the histogram and the driver’s words. Dampers reward disciplined back-to-back testing more than any other component.

What good damping looks like in the data

The acid test is the contact-patch load variation (how steady the vertical tyre load stays over a bumpy section). Minimising it is the entire point of damping and the reason strong teams quote it. A 4-post or 7-post rig does this in the lab by shaking the car and measuring tyre load; on-track you approximate it from pushrod strain gauges or from damper-velocity + spring rate. Lower variation = steadier grip = faster, more confident car.

DO	DON'T
✓ Tune dampers only after springs/bars are set.	✗ Don't use dampers to mask a wrong spring rate or bad geometry.
✓ Use damper-velocity histograms, not guesswork, to judge settings.	✗ Don't run huge rebound — the car jacks down and loses travel.
✓ Run rebound slightly higher than compression as a start, then test.	✗ Don't set dampers by feel alone on an amateur driver — corroborate with data.
✓ Change one adjuster at a time and log back-to-back laps.	✗ Don't forget cold dampers read differently — note oil temperature.
✓ Buy dampers with independent low/high-speed and bump/rebound adjustment if budget allows (e.g. the quality MTB/kart dampers FSAE teams use).	✗ Don't ignore that MR shapes damper velocity too — a rising-rate rocker changes effective damping through travel.

Part H — The coupling map: how every decision pulls three others

This is the part you asked for most directly: nothing in vehicle dynamics changes alone. Every decision propagates. A team that does not see the propagation fixes one problem and creates two. Here is the map — first as a table you can keep open while designing, then as a few worked chains.

The dependency table

Table H-1. The coupling map. Read the third column before committing the first — that is the whole discipline.

If you change...	Direct effect	Knock-on effects (watch these)
Lower CoG height	Less load transfer (long & lat)	More total grip; less dive/squat/roll; usually all-positive — chase it hard
Wider track	Less lateral load transfer	More grip; but more frontal area/drag, worse packaging, tighter-corner line changes, heavier arms
Longer wheelbase	Less longitudinal transfer; more stable	Lazier yaw (more understeer-ish), larger turning radius, heavier/stiffer chassis needed
Lighter unsprung mass	Better bump following; less yaw inertia	Steadier contact-patch load → more grip; quicker response — but costs money (Ti, CF, 10" wheels)
Stiffer front ARB	More front roll stiffness	Higher front TLLTD → more understeer; less front mechanical grip over bumps
Stiffer rear ARB	More rear roll stiffness	Higher rear TLLTD → more oversteer; watch snap on a rear-heavy car
Stiffer springs (both ends)	Less roll & ride motion	Sharper turn-in, better aero platform — but LESS mechanical grip on bumps; harsher
Higher roll centre	More geometric (instant) transfer; less roll	Faster response but jacking risk, harshness, mid-corner balance shift if it migrates
More static negative camber	Better grip in hard corners	Worse braking/traction grip (patch tilted straight-line), inner-edge overheating, more tyre wear
More caster	More self-centring & corner camber	Heavier steering; more driver fatigue; more dynamic camber (can help front grip)
More front Ackermann	Less inside-tyre scrub low speed	Can be wrong at high load where tyre wants different slip — tie to tyre data
More rebound damping	Better settle control	Car can jack down over bumps → lost travel, lost grip
Higher tyre pressure	Sharper response, less rolling drag	Smaller patch, centre overheats, peak grip can drop (Book One Part IX)
More front brake bias	Later front lock-up margin	Rear can lock first if over-shifted; interacts with longitudinal load transfer
More downforce	More grip at speed	More drag; balance shifts with speed (CoP vs CoG); needs stiffer springs to hold ride height

Three worked chains (follow the dominoes)

Chain 1 — A “fix” for mid-corner understeer that creates three new problems if you are not careful:

Driver says it pushes mid-corner → you stiffen the front ARB to ... wait, that adds front load transfer and makes it push MORE. The right move is SOFTEN the front bar (or stiffen the rear) to lower front TLLTD. But softening the front bar also lets the front roll more → front camber goes positive on the outer tyre → you may now need more front static camber or more camber gain. And more rear bar raises rear TLLTD → watch for exit oversteer on this 45/55 car. One balance change, three things to check: camber, exit stability, and that you did not just trade understeer for a loose rear.

Chain 2 — Chasing low roll the wrong way:

You want less roll, so you raise the roll centre → less roll, yes → but now the car jacks up mid-corner (CoG rises, grip falls) AND if the arms are short the RC migrates, so balance changes through the corner → the driver reports inconsistency. The better path to “less roll” was a stiffer ARB (roll only) or accepting the roll and fixing camber. Geometry changes are permanent and couple hard — reach for the bar first.

Chain 3 — The unsprung-mass investment paying off twice:

You spend on 10” wheels and lighter uprights → unsprung mass drops → (a) the tyre follows bumps better, so contact-patch load is steadier and average grip rises, AND (b) mass moves off the car’s extremities, so yaw inertia drops and the car changes direction more eagerly. One decision, two independent gains — which is why strong teams lead their VD section with unsprung mass. This is the rare chain where the dominoes all fall the right way.

How to use this part in practice

Before any change, say out loud: “this changes [direct], which will also change [X] and [Y], and I will verify [X] and [Y] with [measurement].” If you cannot name the knock-ons, you are not ready to make the change. This habit is the difference between a team that converges on a good car and one that chases its tail all season.

Part I — The design sequence: what to decide first

Because everything couples, ORDER matters. Decide the things that constrain the most other things first, and the things you can trim later, later. Here is a sequence that works for a from-scratch FSAE car.

1. **1. Set performance targets** — what does the car need to do? Target lateral g, braking g, autocross-dominated vs endurance, driver skill level. These set everything downstream. Write them down and defend them in the design event.
2. **2. Fix mass targets** — total mass budget, weight distribution, CoG height target, yaw-inertia goal (mass centralisation). These feed every load-transfer and balance calculation.
3. **3. Choose the tyre** — from the tyre study ([Book One](#)): size, compound, operating loads, peak μ , cornering stiffness, load sensitivity. The tyre decides the grip you are designing the rest of the car to deliver to it. Do not design suspension before you know the tyre.
4. **4. Set the footprint** — track width and wheelbase, from the load-transfer/packaging/turning-radius trade (Parts C, E). Hard to change later.
5. **5. Design the kinematics** — camber curve, roll centres/roll axis, caster/KPI/scrub, Ackermann, bump steer (near zero), anti-geometry. These are **HARDPOINTS** — permanent once welded. Spend the most CAE time here.
6. **6. Size springs & bars** — ride-frequency targets → wheel rates → springs through motion ratio; roll-gradient target → total roll stiffness; TLLTD → ARB sizing. These you can trim at the track, so set sensible and move on.
7. **7. Choose dampers** — from damping-ratio targets; refine on track with histograms. The last thing, because it tunes what everything else set up.
8. **8. Instrument it** — plan the sensors and logging NOW (Parts J–L), not after the car is built — you need the data to prove every number above. Instrumented from day one beats bolted-on later.
9. **9. Validate & correlate** — corner-weight, measure CoG and roll gradient, run skidpad and step-steer, correlate to the sim, iterate. The loop that turns a design into a fast car.

The iteration reality

This is not a one-pass waterfall. Step 3 (tyre) may force you back to step 2 (mass) if the good tyre is heavy; step 5 (kinematics) may reveal a packaging clash that moves step 4 (track). Loop deliberately, but freeze the hardpoints (steps 4–5) as early as you responsibly can — every week you leave geometry “open” is a week the chassis and uprights teams cannot finalise. Design is managed convergence, not endless openness.

Part J — Telemetry I: what to measure, the sensors, the architecture

Telemetry (data acquisition, “DAQ”) is how vehicle dynamics stops being opinion and becomes measurement. Every number in Parts A–I — CoG, roll gradient, TLLTD, damping, balance — can and should be verified on the car. This is also where a typical FSAE electronics stack (e.g. a Teensy/ESP32 logging over CAN, a 9-DOF IMU, a wireless link) becomes a vehicle-dynamics tool, not just a dashboard. The definitive reference for everything in Parts J–L is Segers, *Analysis Techniques for Racecar Data Acquisition* — treat it as the companion textbook to these three chapters.

What to measure, and why each channel earns its place

Table J-1. The FSAE DAQ channel list, ranked by VD value. If budget is tight, prioritise the top five: shock pots, IMU, wheel speed, steering, brake pressure.

Channel	Sensor	What it gives you (the VD payoff)
Damper position $\times 4$	Linear potentiometer (shock pot)	Ride height, roll, pitch, heave, and damper velocity (differentiate) — the backbone of VD data
Lateral/long/vert accel	IMU accelerometers	Load transfer, g-g diagram, combined-g usage
Yaw/roll/pitch rate	IMU gyroscopes	Yaw response, stability, rotation vs steer
Wheel speed $\times 4$	Hall/inductive + toothed ring	Speed, slip ratio, lock-up/spin, mini-locking
Steering angle	Rotary encoder/pot on column	Driver input; understeer gradient; Ackermann check
Brake pressure $\times 2$ (F/R)	Pressure transducer in line	Brake bias, threshold braking, trail-brake behaviour
Throttle position	APPS/TPS	Driver input; on/off-throttle balance; traction events
Pushrod/damper load	Strain gauge (load cell) on pushrod	ACTUAL tyre vertical load $\rightarrow$ contact-patch load variation, real load transfer, MR verification
Tyre temperature	IR sensor per tyre (or pyrometer in pits)	Camber/pressure diagnosis, thermal window (Book One Part IX)
Tyre pressure	TPMS	Hot pressure tracking
Position & speed	GPS (10–25 Hz)	Track map, lap/segment times, corner ID, speed trace
Engine	ECU over CAN	RPM, lambda, coolant, gear — context and traction events

If you can add only ONE advanced channel, add this

Strain-gauged pushrods (turning each pushrod into a load cell) give you the ACTUAL vertical load at each tyre, in real time. That single upgrade lets you measure true load transfer, verify your motion ratio, and quantify contact-patch load variation directly — the exact metric strong teams argue about. It is the highest-value sensor most FSAE teams skip.

The system architecture

A modern FSAE DAQ is a small distributed network, which maps directly onto what you already have:

1. **Sensors** → each sensor outputs a voltage, frequency, or digital signal. Analog sensors (pots, pressure, strain) need conditioning: amplification (especially strain — microvolts), filtering, and scaling to the ADC range.
2. **Acquisition nodes** → microcontroller nodes (your Teensy 4.1 for fast chassis/VD channels, ESP32 for engine) read the conditioned signals via their ADCs and timers, timestamp them, and publish them.
3. **CAN bus** → all nodes share one CAN 2.0B bus at 500 kbps (yours). CAN is the right choice: robust, noise-tolerant, multi-node, deterministic. Each message carries a few channels with an ID and a timestamp.
4. **Log + stream** → a logger writes every CAN message to an SD card (the permanent record), and a wireless link (your nRF24L01, or LoRa/Wi-Fi/XBee for more range) streams a subset to the pit for live monitoring.
5. **Base station** → a laptop in the pit decodes the stream live (alarms: temps, pressures, voltages) and, after the run, the full SD log is pulled into analysis software (MoTeC i2, AiM RaceStudio, or your own MATLAB/Python).

The reason this architecture wins: it is modular (add a sensor = add a node or a message, no rewiring the car), noise-tolerant (CAN differential signalling survives the electrically filthy environment near an ignition system), and time-aligned (one clock domain so channels can be cross-plotted). Your existing electronics already implement the hard parts — Parts K and L are about using them for VD.

Part K — Telemetry II: implementation in detail

This is the “exact ways to implement it” you asked for. Get these right and your data is trustworthy; get them wrong and you will confidently draw the wrong conclusions, which is worse than no data.

Sampling rates and anti-aliasing (the part teams get wrong)

Every channel must be sampled fast enough to capture the motion it represents — at least twice the highest frequency of interest (the Nyquist rule), and in practice 5–10× for clean signals. Sample too slowly and fast events alias into fake low-frequency wiggles you cannot remove later.

Table K-1. Sampling rates. Rule: anything you will DIFFERENTIATE (position→velocity) needs the highest rate; anything slow (temps) can be low.

Channel	Suggested rate	Why
Damper position	200–1000 Hz	You differentiate it for velocity — needs headroom; bump events are fast
IMU accel & gyro	100–500 Hz	Transients, yaw response, kerb strikes
Wheel speed	100–500 Hz (or timer capture)	Lock-up and wheelspin happen fast
Steering angle	100–200 Hz	Capture quick inputs and step-steer
Brake/throttle	100–200 Hz	Threshold braking, blips
Pushrod load	200–1000 Hz	Contact-patch load variation is high-frequency
Tyre temp / pressure	10–25 Hz	Thermal changes are slow
GPS	10–25 Hz	Position is smooth; fuse with IMU for more

Put a low-pass anti-aliasing filter (analog RC, or the sensor’s built-in) BEFORE the ADC on fast analog channels, cutting just below half the sample rate. You can always filter more in software; you can never un-alias.

Sensor selection, mounting and calibration — channel by channel

Shock pots (damper position)

- **Mount** — match the sensor stroke to the damper stroke (plus margin) so you use the full resolution; mount inline with the damper so it reads true shaft travel; protect the body and wiper from debris.
- **Calibrate** — at full droop and full bump (on stands, remove the spring), record the raw counts at known wheel positions; fit counts→mm. Then measure the motion ratio by moving the wheel a known amount and reading the pot — this VERIFIES your CAD MR ([Part E](#)).

IMU (accelerometers + gyros)

- **Mount** — mount rigidly, as near the CoG as practical, axes aligned to the car (x forward, y right, z down). Off-CoG mounting adds rotational artefacts you must correct for; note the offset so you can.
- **Calibrate** — level the car and zero the accelerometers (remove the 1 g offset on the right axis, set lateral/-longitudinal to zero at rest); for gyros, zero the rates at standstill (bias). Re-zero when temperature drifts. A 9-DOF IMU (accel+gyro+mag, like yours) lets you fuse for drift-free attitude.

Wheel speed

- **Mount** — a toothed ring and a Hall/inductive pickup; more teeth = better low-speed resolution. Mount the gap to spec and shield the signal.
- **Calibrate** — roll the car a known distance (or use GPS speed on a straight) to tie pulses→speed via the rolling radius; this also gives you the effective rolling radius you need for slip ratio ([Book One Part VI](#)).

Brake pressure, throttle, steering

- **Mount** — pressure transducers tee into the front and rear circuits (choose a range above your max line pressure); throttle from the APPS/TPS; steering from a rotary sensor on the column or rack.
- **Calibrate** — brake: known master-cylinder pressures or the ratio at a hard push; steering: lock-to-lock at known rack positions, zero on-centre; throttle: closed and WOT endpoints.

Strain-gauged pushrods (load)

- **Mount** — bond gauges in a bridge on the pushrod (temperature-compensated); the pushrod must be in near-pure tension/compression where you gauge it (not bending) — another reason to load rod-ends axially.
- **Calibrate** — dead-weight or press the rod at known loads and fit micro-strain→newtons; then resolve to tyre vertical load through the known installation geometry. This is fiddly but gives you the gold-standard load channel.

CAN bus and logging discipline

- **Bus** — one 500 kbps CAN 2.0B bus, 120Ω termination at BOTH ends, twisted pair, kept away from ignition leads. Budget bus load (sum of message rates × sizes) below ~60–70% so you never drop frames.
- **Message map** — assign fixed CAN IDs per message, pack related channels together, and WRITE A DBC FILE (the database that maps IDs/bytes→named, scaled channels). The DBC is what lets any tool decode your log — without it, your data is unreadable next year.
- **Time sync** — every node timestamps to one clock (sync off a periodic CAN message) so channels cross-plot correctly; a 1 ms skew ruins a damper-velocity vs load cross-plot.
- **Logging** — log EVERYTHING to SD (the stream is only a subset); name files with date/session/driver; and record a metadata header each run: setup sheet, tyres & heat cycles, ambient/track temp, fuel. Data without context is noise.

Electrical noise — build on good wiring practice

A good FSAE harness already uses PTFE wire, IP68 Deutsch connectors, braided sleeving, star grounding at the battery, and resettable fuses — exactly the right foundation for clean DAQ. The additions for data quality: shield and twist low-level signals (strain, pots), keep sensor grounds separate from power grounds until the single star point, route signal looms away from the ignition and the stator/charging leads, and add anti-aliasing filtering at the ADC. Clean wiring is not just reliability — it is data accuracy.

Part L — Telemetry III: from raw data to answers

Data you do not analyse is just a full SD card. This chapter is a set of recipes — each takes specific channels and produces a specific vehicle-dynamics answer, with the method, what “good” looks like, and the action it drives. Work through them with your own logs and you will have closed the loop from sensor to setup.

Recipe 1 — Ride heights and the aero/mechanical platform

Channels: four shock pots (+ motion ratio).

Method: convert each damper position to wheel vertical position, then to ride height at each corner. Average the fronts and rears for front/rear ride height; the difference is rake.

Output & action: watch how ride height changes with speed (aero) and under braking/acceleration (pitch). Excess front dive under braking that upsets the car → consider spring/anti-dive/bump-damping. Platform wandering at speed → springs too soft for the aero load.

Recipe 2 — Roll angle and roll gradient (verify your roll stiffness & CoG)

Channels: left/right shock pots at an axle (or IMU roll angle).

```
roll angle  $\approx$  arctan( (z_L - z_R) / track ) ; roll gradient = roll  
angle / lateral g
```

where z_L , z_R = left/right wheel vertical positions from the pots; plot roll angle vs measured lateral g; the SLOPE is the roll gradient (deg/g).

Output & action: compare the measured roll gradient to your design target (Part F). If the car rolls more than designed, your real roll stiffness is lower (soft bar, compliant mounts) or your CoG/roll-arm is higher than CAD — both worth knowing. This single plot validates a huge chunk of Parts B, C and F.

Recipe 3 — Damper-velocity histogram (tune the dampers)

Channels: shock pot, differentiated to velocity.

Method: differentiate position to get damper velocity; build a histogram of time spent at each velocity, split bump/rebound and low/high speed (Part G).

Output & action: a healthy histogram is smooth and bell-shaped, centred near zero, not spiked or clipped. A tall spike at zero = too much low-speed damping (car is “stuck”). Long tails with the wheel spending lots of time at high velocity = under-damped over bumps. Adjust the matching adjuster and re-log back-to-back.

Recipe 4 — Understeer gradient (measure your balance)

Channels: steering angle, lateral g (IMU), speed.

Method: the classic constant-radius skidpad test — drive a fixed-radius circle at increasing speed; plot steering angle (corrected for the Ackermann/ratio) against lateral g. Alternatively, derive it from a full lap with the right maths.

```
rising steer-vs-g slope  $\rightarrow$  understeer ; flat  $\rightarrow$  neutral ; falling  $\rightarrow$   
oversteer
```

where The slope IS the understeer gradient K (deg/g). This is the direct, on-car measurement of the balance you computed in Part D.

Output & action: if the measured K disagrees with your sim, your tyre C_α or load-transfer split is off — correlate and fix the model, then trust it. This is the most important handling measurement you can make.

Recipe 5 — Yaw response and stability

Channels: yaw rate (gyro), steering angle, speed.

Method: compare actual yaw rate to the “reference” yaw rate a neutral car would have for that steer and speed; and look at yaw-rate response time and overshoot after a step-steer input (**Part O** test).

Output & action: slow, over-damped yaw response → too much understeer / too much yaw inertia; twitchy, overshooting yaw → toward oversteer / unstable. This is where your yaw-inertia work (**Part B**) shows up in the data.

Recipe 6 — The g-g diagram (how much tyre the car and driver use)

Channels: lateral g and longitudinal g (IMU).

Method: scatter-plot longitudinal g (y) against lateral g (x) for a whole lap. The outer envelope is the friction ellipse the car achieved (**Book One Part VII**).

Output & action: gaps where the cloud does not reach the edge = grip left on the table. A notch between hard braking and hard cornering = the driver is not trail-braking (separating the two) — coaching. A smaller envelope than your tyre data predicts = setup or tyre-temperature problem. This one chart links tyre, car and driver.

Recipe 7 — Contact-patch load variation (validate unsprung mass & damping)

Channels: strain-gauged pushrod load (best) or spring rate × damper position (approx).

Method: over a bumpy section, compute the statistical variation (standard deviation / RMS) of the vertical tyre load about its mean.

Output & action: lower variation = the tyre stays more consistently loaded = more average grip. This is the exact metric strong teams quote (e.g. “RMS contact-patch force variation reduced 9%”). Use it to prove the value of lighter unsprung mass and of a damper change — with a number, not a claim.

Recipe 8 — Slip ratio, lock-up and wheelspin (traction & braking)

Channels: four wheel speeds, GPS/true speed, brake pressure, throttle.

$$\kappa = (\Omega \cdot R_e - V) / V \text{ (per wheel)}$$

where Compare each driven/braked wheel’s surface speed to true speed. κ spiking negative under braking = lock-up; spiking positive on throttle = wheelspin.

Output & action: find where wheels lock (brake bias too far that end) or spin (traction limit / diff behaviour — your LSD). Tune brake bias and diff from this, and feed the real operating slip ratios back to your tyre model.

Recipe 9 — On-car bump steer & motion-ratio check

Channels: shock pots, steering (fixed), toe measurement on stands.

Method: with the steering clamped, cycle the suspension through travel and measure toe change (bump steer) and spring/damper travel vs wheel travel (motion ratio).

Output & action: confirm bump steer is near zero (**Part E**) and that the built motion ratio matches CAD — because every wheel-rate, roll-stiffness and damping number depends on the real MR. Build tolerances mean the as-welded car is never exactly the model; this is how you catch it.

Recipe 10 — Lap/segment time and driver overlays

Channels: GPS + all of the above.

Method: split the track into segments; overlay two laps or two drivers' speed, throttle, brake and steering traces.

Output & action: find WHERE time is lost (entry, apex, exit) and WHY (late braking vs early throttle vs mid-corner speed), which tells you whether the fix is setup (balance in that corner type) or driver technique. This is how telemetry turns into lap time.

The correlation loop — the whole point of Parts J–L

For every recipe, the real win is comparing MEASURED to PREDICTED: measured roll gradient vs sim, measured understeer gradient vs sim, measured g-g vs tyre-data ceiling, measured contact-patch variation vs 4-post prediction. Each comparison either validates your model (now you can trust it to design next year's car) or exposes an error (now you can fix it). A team that closes this loop even on three or four channels is operating at a completely different level from one that just "logs data." That loop, run every test day, is what a vehicle-dynamics program actually is.

Part M — Setup: every adjustment, its effect, and a tuning decision tree

Setup is applying everything above at the track with the tools you built adjustability for (Part I). The goal is a repeatable process, not magic: know what each knob does, change one at a time, measure the result.

The setup toolbox — what each adjustment does

Table M-1. The setup toolbox. “TLLTD” = total lateral load transfer distribution (Parts C–D): the master balance concept behind the spring/bar rows.

Adjustment	Increase it → effect	Mainly use to...
Front tyre pressure	Sharper front, less front grip past optimum	Trim front response & front temps
Rear tyre pressure	Sharper rear, less rear grip past optimum	Trim rear grip & temps
Front camber (more neg)	More front grip in corners, less braking grip	Add mid-corner front grip; fix outer-edge temps
Rear camber (more neg)	More rear corner grip, less traction grip	Add rear corner grip
Front toe-out	Sharper turn-in, more front scrub/heat	Improve initial response
Rear toe-in	More stability	Calm a loose rear
Caster	More self-centring & corner camber, heavier steer	Add feel and front corner grip
Front spring/ARB	More understeer (higher front TLLTD)	Reduce oversteer / calm the car
Rear spring/ARB	More oversteer (higher rear TLLTD)	Reduce understeer / add rotation
Ride height / rake	Platform & (with aero) balance	Set aero platform & CoG
Low-speed bump/rebound	Transient (entry/exit) balance & platform	Trim entry/exit without changing steady balance
High-speed damping	Grip over bumps/kerbs	Keep tyres planted on rough sections
Cross weight (corner wt)	Diagonal balance (L vs R turns)	Even out a car that turns better one way
Diff (LSD) preload/ramps	Entry vs exit rotation/traction	Tune power-on/off balance & traction
Brake bias	Front vs rear lock-up	Set threshold braking & entry stability

The balance-tuning decision tree

Diagnose balance by WHERE in the corner it happens — entry, mid, or exit — because each points to different tools. Ask the driver “where does it push or get loose?” and use this:

Table M-2. Balance decision tree. Dampers for entry/exit (transient); ARBs/springs for mid-corner (steady); diff & brakes for power-on/off and braking phases.

Symptom	Most likely cause	First fixes to try (one at a time)
Entry understeer (won’t turn in)	Front not taking load fast enough; too much front; cold fronts	Less front low-speed compression/rebound; less front ARB; more front camber; front pressure; more front toe-out
Mid understeer (pushes at apex)	Steady front TLLTD too high; front grip-limited	Soften front ARB / stiffen rear; more front camber; check front tyre temps/pressure

Symptom	Most likely cause	First fixes to try (one at a time)
Exit understeer (won't rotate on power)	Diff too locked; rear squatting; power understeer	Reduce diff power-ramp/preload; a touch more rear rebound; check traction/wheelspin
Entry oversteer (loose on brakes)	Rear unloading; too much rear brake; trail-brake	Move brake bias forward; more rear rebound control; soften rear entry damping
Mid oversteer (loose at apex)	Rear TLLTD too high; rear grip-limited	Soften rear ARB / stiffen front; more rear camber; rear pressure; rear toe-in
Exit oversteer (steps out on power)	Too much rear load transfer on power; diff; power	Diff ramp/preload; gentler throttle map; small rear spring change; check rear temps
Inconsistent / changes each lap	Roll-centre migration; temps/pressures moving; compliance	Check RC migration (geometry); stabilise temps/pressures; check for flexing mounts/rod-ends

Corner weighting and the baseline

- **Corner-weight (set up pad)** — on a flat, level pad with the driver aboard and target fuel, adjust the spring perches (ride-height collars) so the diagonal (cross) weight is ~50%. Uneven cross weight makes the car turn better one way than the other — a hidden handful of points on a symmetric autocross.
- **Baseline + setup sheet** — record a known-good setup you can ALWAYS return to. After any test session that went sideways, reset to baseline rather than stacking guesses. A setup sheet (every adjustable value + tyre/temp/fuel state) is how you keep the car reproducible and how you learn across sessions.

The one discipline that beats talent

Change ONE thing, drive the SAME corner, compare the DATA and the driver's words, write it down. Teams lose whole weekends by changing three things at once and never knowing which helped. Slow, logged, one-variable testing converges; heroic multi-change guessing does not.

Part N — Design dos & don'ts, and the classic FSAE mistakes

Everything above, distilled into the hard rules and the traps that sink rookie VD teams. Read this before every design review.

Geometry & hardpoints

DO	DON'T
✓ Freeze hardpoints early — chassis and uprights depend on them.	✗ Don't chase "less roll" with a high roll centre — jacking.
✓ Design bump steer to ~zero and verify on the built car.	✗ Don't run rear toe-out, ever.
✓ Keep roll centres low and their migration small.	✗ Don't use maximum camber gain and expect a stable roll centre — you can't have both.
✓ Prioritise lateral geometry (camber, RC, bump steer) over anti-dive/squat.	✗ Don't put a spherical bearing where it sees bending moments.
✓ Load rod-ends and spherical bearings axially, never in bending.	✗ Don't finalise geometry without a kinematics sim AND a plan to measure it.

Springs, bars & dampers

DO	DON'T
✓ Choose springs by ride frequency; set balance with bars.	✗ Don't stiffen everything to "feel racy" — you lose mechanical grip.
✓ Run the softest rates that control ride height, roll & aero.	✗ Don't forget MR^2 when converting spring to wheel rate.
✓ Make at least one ARB and the dampers adjustable.	✗ Don't run excessive rebound (jacking down).
✓ Verify motion ratio and roll gradient on the car.	✗ Don't set balance with dampers (they're transient, not steady-state).
✓ Tune dampers with velocity histograms, one click at a time.	✗ Don't trust CAD masses — corner-weigh.

Uprights, hubs, bearings (where VD meets structures)

DO	DON'T
✓ Size wheel bearings for COMBINED radial + axial (cornering) load — your move to two tapered rollers per wheel is exactly right.	✗ Don't keep a single ball bearing that can't take axial corner loads (a classic failure).
✓ Use a dynamic magnification factor on load cases to cover transients.	✗ Don't optimise upright mass so hard it becomes compliant — compliance steals your geometry.
✓ Target lightest unsprung mass that meets stiffness & life (7075/Ti where it pays).	✗ Don't ignore bearing preload — it changes stiffness and life.
✓ Check upright/hub stiffness, not just strength — a flexy upright adds compliance camber.	✗ Don't forget brake/thermal loads into the upright.
✓ FMEA the single-failure-point parts (upright, hub, pushrod, steering).	✗ Don't design parts you can't inspect or replace between sessions.

The classic FSAE VD mistakes (learn them cheaply, here)

- **Sim with no correlation** — a beautiful Lotus/OptimumKinematics model and no sensors to check it. The sim is a hypothesis until data confirms it.

- **Over-stiffening** — stiff springs + big bars to “reduce roll,” then no grip on bumpy autocross. Mechanical grip lives in compliance.
- **False precision** — every number to five significant figures from a sim, zero measured. Judges probe exactly here. Round, and cite the source.
- **Tyre last** — designing suspension before the tyre study is done — you are designing to deliver load to a tyre you haven't characterised.
- **Bump steer ignored** — never built or verified; the car steers itself over bumps and no one knows why it's inconsistent.
- **Compliance leaks** — spherical bearings in bending, pushrods that see side loads, uprights that flex — compliance that quietly undoes the geometry you designed.
- **No baseline/logbook** — nobody logged the setup, tyres, temps and heat cycles, so last month's fast setup is unrecoverable. The logbook is the team's memory.

Part O — Process, CAE and validation

The workflow that ties design to proof — and the one the design event scores.

Targets → requirements → design → test (the V)

Start from performance targets (**Part I**), flow them down to subsystem requirements (roll gradient, camber targets, bearing loads), design to those, then test UP the other side of the V: component tests validate parts, vehicle tests validate the system against the original targets. Every requirement should have a matching test.

CAE — the tools and what each is for

- **Multibody geometry** — kinematics/compliance — camber curves, roll centres, bump steer, motion ratio, anti-geometry. Lotus Shark/Suspension, OptimumKinematics, Adams/Car. This is where your hardpoints are born and checked.
- **Lap & handling sim** — a point-mass or lumped-parameter model that turns tyre + mass + geometry into lap time and lets you weight design choices by points (OptimumLap, or your own). A “lumped-parameter model” and tools like OptimumLap live here.
- **FEA (structures)** — uprights, hubs, bell cranks, A-arms under cornering/braking/accel load cases WITH a dynamic magnification factor; check stiffness as well as stress; FMEA the critical parts. This is the right process.

Physical validation tests (run these)

Table O-1. The validation test matrix. Each closes a loop between a design number and reality (**Part L** shows how to read them).

Test	Measures	Validates
Corner-weighting (pad)	Static corner & cross weights	Mass model, ride heights, symmetry
CoG tilt test	CoG height	Load-transfer model (Parts B–C)
Skidpad (constant radius)	Lateral g, understeer gradient	Tyre grip & balance (Parts D, L)
Step-steer	Yaw response, settling time	Transient model, yaw inertia (Parts B, L)
4-post / 7-post rig	Contact-patch load variation	Springs, dampers, unsprung mass (Parts F, G, L)
Straight-line brake	Decel g, lock-up, bias	Brake & longitudinal-transfer model
Bump-steer / MR on stands	Toe vs travel, motion ratio	Kinematics as-built (Part E)

What “good” looks like to a design judge

Not the fanciest sim — the tightest loop. “We targeted X, designed for X, measured X on the car, and it matched within Y% (or it didn’t, and here’s what we learned)” beats any amount of unvalidated analysis. Bring one fully-closed loop — say, measured vs predicted understeer gradient or roll gradient — and you will out-present teams with prettier CAD.

BOOK THREE

| FORMULA STUDENT VEHICLE DYNAMICS

The Build Playbook

Design to fabricate to test to validate: ten phases, decision gates, procurement timing, and the sub-system dependency map.

How to use this playbook

Books 1 and 2 taught the why and the how-it-couples. This book is the checklist you actually execute. It is written so a new team member who has read Books 1–2 can start at **Phase 0** and, by following the steps in order, arrive at a scrutineering-ready, validated suspension — knowing at every step which software to open, what number to produce, what the number should be, and exactly what to change if it is wrong.

How the steps are written:

- **Phases & gates** — the work is split into ten phases (0–9), each ending in a decision gate you must pass before spending money or cutting material on the next phase.
- **Steps & tables** — inside each phase, numbered steps say what to do. The concrete engineering checks are in “analysis tables” with four columns: the check, the starting target, the pass criterion, and — crucially — what to change if you are outside target. That last column is the “based on the value, correct this” you asked for.
- **Boxes** — amber boxes are STOP-gates. Rust boxes tell you when to order parts. Teal boxes are judgment/points advice.

The numbers in this book are STARTING targets, not laws

Every target value here (camber angles, roll gradient, factors of safety, frequencies) is a sensible FSAE starting point to get you moving. Your own lap-sim (**Phase 2**) and your tyre data (**Phase 1, Book One**) refine them into YOUR car’s targets. Use these to begin and to sanity-check; replace them with your validated numbers as you generate them. And every rule-driven number (wheelbase, track, travel) must be checked against the current FSCC/FSAE rulebook for your competition year — rules change.

The two golden rules of this whole process

If you remember nothing else:

- **1. Respect the order** — tyre first, then concept, then geometry, then springs/dampers, then build, then test. Each phase hands the next its frozen inputs. Jumping ahead means redesigning later — the most expensive mistake a team makes.
- **2. Freeze deliberately** — hardpoints (**Phase 3**) and part designs (**Phase 4**) get FROZEN at a gate. After freeze, you do not quietly “tweak” a pickup point — chassis, uprights and procurement all depend on it. Change after freeze only through a logged change-review. Unfrozen geometry is why cars miss competition.

The master timeline at a glance

A typical ~8-month (34-week) build for a team with no carry-over car. Compress or stretch to your own calendar, but keep the ORDER and the gates. Phases overlap on purpose — the arrows below show where one feeds the next.

Table 0-1. The ten phases. Note the overlap: order parts (*Phase 4*) while finishing geometry, build the DAQ while fabricating, etc.

Phase	Weeks (typical)	What you produce	Exit gate
0 — Mobilise	0–2	Targets, fixed assumptions, toolchain, team, trackers	G0 Targets signed off
1 — Tyre selection & model	1–4	Chosen tyre + Magic-Formula model + operating metrics	G1 Tyre locked
2 — Concept & lap-sim	3–6	Frozen vehicle targets (WB, track, mass, CoG, freqs)	G2 Concept freeze
3 — Kinematics / hardpoints	5–11	Validated geometry; frozen hardpoints	G3 Hardpoint freeze
4 — Load cases & component design	9–16	FEA-validated parts; drawings; BOM	G4 Design freeze
5 — Springs/dampers/ARB	12–17	Spring rates, damper spec, ARB sizes	G5 Rates frozen
6 — Fabricate, assemble, QC	15–24	Built, inspected corners on the car; DAQ harness	G6 Rolling chassis
7 — Shakedown & baseline	23–26	Corner-weighted, measured car; first laps	G7 Reliable runner
8 — Test & validate	25–32	Correlated model; met performance targets	G8 Performance signed off
9 — Competition readiness	30–34	Scrutineering-passed; setup sheets; spares; drivers	G9 Competition ready

How a gate works

At each gate, the subsystem lead reviews the exit deliverables against the pass criteria in that phase. If every item passes, the gate is signed and the next phase's irreversible actions (spending money, cutting material, welding) are authorised. If anything fails, you loop within the phase — you do NOT carry an open problem across a gate. Write the sign-off down (date, who, which version of the numbers). This single discipline is what separates teams that finish from teams that thrash.

The subsystem dependency map: what to fix before what

You asked for the design timeline of each area — tyres, suspension, steering and their components — set against each other by dependency. This is that map, and it is the backbone of the whole playbook. Vehicle dynamics is one connected system, so the ORDER you freeze things in matters as much as the values: fix an upstream decision too late and three sub-teams sit idle; finalise a downstream part too early and you rebuild it when the upstream number moves. The table shows, for each decision, what must be fixed BEFORE it and what it UNBLOCKS once fixed.

Table D-0. The master dependency map. Read a row as: this cannot start until the middle column is fixed; once it is fixed, the right column can start. The Phase column is when it is frozen.

Decision / parameter	Can't be fixed until... (depends on)	Once fixed, it unblocks	Phase
Performance & reliability targets	Rules + team intent	Literally everything downstream	0
Tyre size & Magic-Formula model	Performance targets	Lap sim, steering ratio & Ackermann, wheel/upright envelope	1
Mass & weight-distribution targets	Targets, rough packaging	Load transfer, lap sim, spring sizing	0–2
CoG height target	Mass, packaging	Load transfer, roll-gradient target	2
Wheelbase & track	Tyre, lap sim, rules	All kinematics, chassis, steering reach	2
Wheel & rim size / offset	Tyre, brake package	Upright envelope, scrub/KPI, bearings	2–3
Suspension hardpoints (camber, RC, anti-)	WB/track, wheel, tyre	Uprights, rockers, chassis tabs, steering arm	3
Steering rack (ratio, travel, position)	Tyre peak-slip band, cockpit packaging	Tie-rod inner point, column, Ackermann	3
Steering arm length & Ackermann	Tyre peak slip, hardpoints, rack	Steering effort, bump steer	3
Tie-rod inner point (bump steer ≈ 0)	Rack position + hardpoints	Geometry freeze (G3)	3
Steering column, U-joints, QR, wheel	Rack + cockpit ergonomics	Steering assembly, scrutineering	3–4
Suspension load cases	Tyre μ , mass, load transfer, hardpoints	ALL component FEA	4
Uprights / hubs / wheel bearings	Hardpoints, wheel, load cases	Corner assembly, procurement	4
Rockers & pushrods (motion ratio)	Hardpoints, load cases	Spring sizing, assembly	4
Spring rates	Ride-freq target + motion ratio (needs rocker)	Roll stiffness, assembly	5
ARB sizing	Roll-gradient target + spring roll share	On-car balance range	5
Dampers	Damping target, loads, motion ratio	Damper tuning	order 4 / tune 8
DAQ sensors & harness	Part geometry (mounts), pushrods	All validation	4–6

The dependency chains that bite teams (read these twice)

Tyre → steering — Your steering ratio and Ackermann are set by the tyre’s peak-slip band and aligning moment (**Book One**). So the rack ratio and steering-arm length cannot be finalised until the tyre model exists (**Phase 1**) — steering depends on tyres, not the reverse. Picking a rack “because we had one” before the tyre study is how teams end up with the wrong ratio.

Hardpoints ↔ steering outer ↔ upright — The outer tie-rod point and the steering arm are PART of the upright, and their position sets both Ackermann and bump steer. So the steering outer geometry is frozen TOGETHER with the suspension hardpoints at G3, and the upright cannot be finalised until then. Steering and suspension geometry are one freeze, not two.

Rack → tie-rod inner → bump steer — Bump steer is the matching of the tie-rod’s arc to the suspension’s arc, so you must choose and POSITION the rack before you can zero bump steer. Rack height/length first, tie-rod inner point second, bump-steer check third — all inside **Phase 3**.

Rocker (motion ratio) → springs — Final spring rates need the real motion ratio, which is set by the rocker — a Phase-4 component. That is why spring sizing (**Phase 5**) deliberately trails component design, and why you verify the motion ratio on the built car before trusting any rate (**Phase 7**).

When each subsystem is frozen

Table D-0b. The freeze order. Tyre first, geometry (suspension AND steering together) next, then components, then rates — then validate.

Subsystem	Frozen at	Critically depends on
Tyre	G1 (Phase 1)	Performance targets
Vehicle concept (WB, track, mass/CoG, freqs)	G2 (Phase 2)	Tyre + lap sim
Suspension kinematics (hardpoints)	G3 (Phase 3)	Concept + wheel + tyre
Steering geometry (rack, arm, Ackermann, bump steer)	G3 (Phase 3)	Tyre peak-slip + hardpoints + rack
Structural components (upright/hub/bearing/rocker/pushrod/steering)	G4 (Phase 4)	Frozen geometry + load cases
Springs / ARB / dampers	G5 (Phase 5)	Ride & roll targets + motion ratio
All of the above, validated	G8 (Phase 8)	Built car + telemetry

Steering is not a late add-on

The most common steering mistake in FSAE is treating it as a bolt-on after the suspension is “done.” It is not — its outer points live on the upright, its ratio comes from the tyre, and its bump steer is a suspension-arc problem. Design steering INSIDE Phases 1 (ratio input), 3 (geometry) and 4 (components), on the same freeze schedule as the suspension. A team that does this never discovers at assembly that the rack won’t package or the Ackermann is wrong.

Phase 0 — Mobilise: targets, assumptions, toolchain (Weeks 0–2)

You cannot design anything until you have decided what the car must do and fixed the handful of numbers everything else depends on. This phase produces no parts — it produces the agreements that make the rest possible.

Step 0.1 — Read the rulebook and extract the hard constraints

Open the current FSCC/FSAE rulebook for your competition year and list every rule that touches suspension, steering, wheels and ride height. These are non-negotiable boundaries on the whole design. The classic ones (VERIFY the current numbers yourself):

Table 0-2. Rule-driven constraints to extract FIRST. These are inputs to *Phase 2*, not suggestions.

Rule area	Typical requirement (verify current year)	Impacts
Wheelbase	≥ 1525 mm	WB choice, <i>Phase 2</i>
Track	Smaller track $\geq 75\%$ of the larger	Front/rear track choice
Suspension travel	≥ 50.8 mm usable (≥ 25.4 jounce + 25.4 rebound) with driver	Spring/motion-ratio/geometry travel
Functional suspension	Dampers front & rear, working, with the driver aboard	You cannot run it solid
Ground clearance	No part (other than tyres) touches ground during dynamic events	Ride height, undertray
Wheels/tyres	Rim size, two tyre sets (dry/wet) rules	Tyre & wheel choice
Steering	Free play limit, positive stops, no rack-and-pinion disengagement	Steering design
Tilt table	No fluid leaks and no rollover at 60° (driver aboard)	CoG height budget

Step 0.2 — Set vehicle-level performance targets

Decide, as a team, what “good” means in numbers. These drive every later choice and are what you defend in the design event.

- **Performance** — target lateral grip (e.g. ~ 1.4 – 1.6 g mechanical, more with aero), braking (~ 1.4 – 1.6 g), and the accel/skidpad/autocross/endurance balance. An autocross-and-endurance car (FSCC) is cornering-and-reliability dominated — bias decisions that way.
- **Reliability** — reliability is a target too: the car must finish ~ 22 km of endurance without a suspension failure. Design life and inspection-ability are requirements, not afterthoughts.

Step 0.3 — FIX the onset assumptions (the “lock-list”)

This is the most important table in the book. You cannot start geometry or structures until these are fixed as working assumptions. They are allowed to be rough — you will refine mass and CoG as the car is detailed — but SOMETHING must be written down so parallel work can begin. Fix them, label them “assumed vX”, and update under change-control.

Table 0-3. The onset lock-list. Nothing downstream starts until these exist as written assumptions.

Parameter	Starting value to assume	How you'll firm it up
Total mass (w/ driver)	Set a target, e.g. ~230–300 kg	Mass budget per subsystem; weigh as built
Weight distribution	~45–50% front (yours 45/55)	Corner-weigh the real car (Phase 7)
CoG height	Target $\leq$ ~300 mm (yours 270)	Tilt test (Phase 7/8)
Wheelbase	Pick within rules, e.g. 1530–1600 mm	Lap-sim + packaging (Phase 2)
Track (F/R)	e.g. 1150–1250 mm, $F \geq R$	Lap-sim load-transfer study (Phase 2)
Wheel/tyre	Size + 10" vs 13" shortlist	Tyre study (Phase 1)
Ride height	~30–50 mm	Aero + clearance (Phase 2)
Driver envelope	5th–95th percentile	Ergonomics/seat (fixed early)
Usable wheel travel	$\geq$ rule minimum + margin (e.g. ± 30 mm)	Geometry + spring choice (Phase 3,5)
Aero package	Decide: undertray/wings/none	Aero team; affects springs/platform

❏ Procurement thinking starts NOW (even though you order nothing yet)

Identify the long-lead items today and start the paperwork/quotes: dampers (weeks to months), wheels and tyres, TTC data access, billet 7075 / 4130 tube, wheel bearings, rod ends, and the steering rack. You will place orders in Phases 3–5, but suppliers, quotes, import/customs time (relevant in Bangladesh/for FSCC) and budget approvals must be lined up now or they become your critical path. Make a procurement tracker: item / spec / supplier / quote / lead time / order-by date / owner.

Step 0.4 — Stand up the toolchain and the team

Table 0-4. Minimum toolchain. Everything except the commercial MBD/FEA can be done free (OptimumKinematics/Lotus student, OptimumLap, MFeval).

Job	Tool (pick one per row)	Used in phase
Tyre data & model	MATLAB + your tool / MFeval / OptimumTire	1
Lap & handling sim	OptimumLap (point-mass) → later a full MBD	2, 8
Suspension kinematics	Adams/Car, OptimumKinematics, or Lotus Shark	3
CAD	SolidWorks / CATIA / Fusion	3–6
FEA (structures)	ANSYS / SolidWorks Simulation	4
Data acquisition	Your Teensy/ESP32 CAN logger + i2/RaceStudio/MATLAB	6–8
Trackers	Target sheet, BOM, procurement & test logs (shared)	all

- **Team** — name a VD/suspension lead, a structures/FEA owner, a manufacturing owner, a DAQ owner, and a procurement owner. Each owns their phase deliverables and gate sign-offs. For a small team one person wears several hats — just make ownership explicit.

🔑 GATE G0 — before any design work

Pass when: the rule constraints are listed; vehicle performance targets are written; the Table 0-3 lock-list is filled with “assumed v1” values and signed off by the team; the toolchain is installed and someone has run a hello-world in each; and the procurement/BOM/target trackers exist. If any lock-list cell is blank, you are not ready to start Phase 1–2 design — fill it, even roughly.

Phase 1 — Tyre selection & model (Weeks 1–4)

The tyre is the foundation (**Book One**). You design the whole car to deliver load to it, so you choose it and model it FIRST. Do not skip to geometry because “we’ll just use last year’s tyre” — at least re-pull its data and its operating-load numbers.

Step-by-step

1. **Get TTC data** — register your university, pay the fee (historically ~USD 500), download the rounds covering your candidate sizes. If you already have access from a past year, confirm it. (**Book One Part XI**.)
2. **Run the tyre tool** — for each candidate size/compound, bin by load/camber/pressure, fit the Magic Formula (your MATLAB tool / MFeval / OptimumTire), apply the friction-scaling factor $\lambda\mu$ (your $k \approx 0.65$). (**Book One Part XII**.)
3. **Extract operating metrics** — read peak μ , cornering stiffness $C\alpha$, load sensitivity and peak slip angle AT YOUR OPERATING LOADS (front $\approx \text{total} \times \% \text{front}/2$, rear similarly), not at the lightest load.

Table 1-1. Tyre model checks. Outputs here are INPUTS to Phases 2–3.

Check	Target (starting)	Pass criteria	If outside target → do this
Tyre model fit	MF fits raw data	Fit R^2 high; curve overlays raw points	Re-bin, check axis/units, refit; drop bad runs
Peak μ (scaled) at op. load	$\approx 1.4\text{--}1.7$	Believable post-scaling	Re-check $\lambda\mu$ source; don't ship raw-belt μ
Candidate comparison	Best μ + manageable load sens.	One size wins on YOUR events	Re-weight by event points; reconsider diameter
Peak slip angle	$\sim 4\text{--}10^\circ$ (load-dep.)	Known at operating load	Feeds steering ratio & Ackermann (Phase 3)
Operating pressure/camber	From sweeps	Best within TESTED range	Don't extrapolate below tested pressure

❑ Order tyres & wheels decision inputs

Lock the tyre size now so the wheel/rim spec (**Phase 3–4**) and the upright/bearing envelope can be designed around it. Order at least one test set of tyres early — you need them for shakedown (**Phase 7**), and compounds/age (heat cycles) matter. Note the two-set (dry/wet) rule if your event enforces it.

🔒 GATE G1 — tyre locked

Pass when: tyre size and compound are chosen with a documented, data-driven reason; you have a validated Magic-Formula model (.TIR or your tool's equivalent); and you have peak μ , $C\alpha$, load sensitivity and peak slip angle AT operating load written into the target sheet. These numbers now feed the lap-sim and the geometry. Do not start hardpoints without them.

Phase 2 — Concept & lap-sim: freeze the vehicle targets (Weeks 3–6)

Now turn “what the car must do” plus the tyre into the top-level numbers the geometry and structures teams need: wheelbase, track, mass/CoG targets, roll gradient and ride frequencies. You do this with a lap/handling simulation so the choices are driven by lap time, not opinion.

Step 2.1 — Build the baseline lap-sim model

1. **Set up** — build a point-mass or simple bicycle model (OptimumLap or your own): mass, weight distribution, CoG height, tyre model from [Phase 1](#), power/gearing, drag/downforce if any, and a representative FSCC autocross/endurance track.
2. **Sanity-check** — confirm it reproduces sane numbers: skidpad lateral $g \approx$ your tyre μ , top speed, lap time in the right ballpark. A sim that disagrees with hand-calcs is wrong — fix before trusting it.

Step 2.2 — Run sensitivity studies to SET the targets

This is the real work of [Phase 2](#): sweep each top-level parameter and watch lap time, to choose values with evidence. Run one variable at a time.

Table 2-1. Concept sensitivity sweeps. Each row ends in a frozen target value with a lap-time reason behind it.

Sweep this	Over roughly	Read	Decision it fixes
Total mass	± 30 kg	Lap time per kg	Mass budget & how hard to chase lightness
Weight distribution	40–55% front	Lap time, balance	Target weight split
CoG height	250–320 mm	Lap time (load transfer)	How hard to chase low CoG
Track width	1100–1300 mm	Lap time vs drag/packaging	Front/rear track
Wheelbase	1530–1650 mm	Lap time, agility	Wheelbase
Grip (μ)	$\pm 10\%$	Lap time per 0.1μ	Shows tyre/setup is king — justifies tyre spend
Downforce (if aero)	0 $\rightarrow$ target	Lap time vs drag	Aero go/no-go & target

Step 2.3 — Choose the architecture

- **Suspension type** — double wishbone, pushrod-actuated inboard dampers (the FSAE default). Decide push vs pull based on packaging/CoG.
- **Wheel size** — typically 10" for low unsprung mass & inertia ([Book Two Part B](#)). Confirm your chosen tyre/rim supports it and brakes package inside.
- **Anti-roll bars** — plan for adjustable ARBs front & rear (your balance knob, [Book Two Parts F, M](#)). Decide now so mounts are designed in.

Step 2.4 — Set the ride & roll targets

Table 2-2. Ride/roll targets. These size springs and bars in [Phase 5](#).

Target	Starting value	Pass criteria	If sim says otherwise →
Ride frequency	2.5–3.5 Hz (non-aero); F slightly < R	Flat-ride, controls travel	Aero car → go stiffer (3.5–5 Hz)
Roll gradient	1.0–2.0 deg/g	Enough compliance, sane camber	Too much roll → more roll stiffness/less CoG arm
Static ride height	30–50 mm	Clears ground under load	Aero platform may force lower + stiffer
TLLTD (balance)	Start ~just forward of neutral	Stable mild understeer	Trim later with ARBs on track

GATE G2 — concept freeze

Pass when the VEHICLE TARGET SHEET is frozen and signed: wheelbase, front/rear track, mass & weight-distribution targets, CoG-height target, ride frequencies, roll-gradient target, ride height, wheel size, and architecture. From here the geometry team designs to these numbers. Changing them after G2 means re-doing **Phase 3** — so get them right now, with lap-sim evidence for each.

Phase 3 — Kinematics & hardpoints (Weeks 5–11)

This is where you place the suspension pickup points so the wheel sees the right camber, roll centre, bump steer and steering behaviour. It is the highest-leverage phase: hardpoints are permanent once welded, and uprights, chassis and rockers are all designed around them. Spend the time here.

Step 3.1 — Set up the kinematic model

In your MBD/kinematics tool (Adams/Car, OptimumKinematics, or Lotus), build one corner from coordinates: wheel centre, upper and lower inner/outer ball joints, tie-rod inner/outer, pushrod ends, and the contact patch. Enter the tyre loaded radius and the track/wheelbase from G2.

How to actually run it (Adams/Car — and the free equivalents)

In Adams/Car: build a double-wishbone SUSPENSION subsystem template with your hardpoints, assemble it into a Suspension Test Rig, then run the three standard analyses — (1) Parallel Wheel Travel (both wheels bump together, ± 30 mm) for camber/toe/roll-centre/half-track vs ride; (2) Opposite Wheel Travel / Roll (wheels move oppositely) for roll-camber and roll-centre migration; (3) Steering (sweep the rack) for bump steer, Ackermann and steer camber. Request outputs: camber angle, toe angle, roll-centre height, half-track change, caster, scrub. OptimumKinematics and Lotus Shark do exactly the same analyses with friendlier UIs — “bump”, “roll” and “steer” sweeps — and are free/cheap for students. Any of the three is fine; the analysis NAMES and OUTPUTS are what matter.

Step 3.2 — Iterate each parameter to target

Run the sweeps, read the curves, and move hardpoints to hit each target. Work roughly top-to-bottom — camber/roll-centre first (they fight each other), then steering, then anti-geometry. Expect several loops.

Table 3-1. The hardpoint tuning matrix — the core of this book. Read each curve from the named analysis; if outside target, make the change in the last column; re-run; repeat until all pass together.

Check (and which analysis)	Target (starting)	Pass criteria	If outside target → move this
Static camber (setup)	-1° to -3° front; $\sim -1^\circ$ to -2° rear	Set in model	Set at the outer ball-joint / shim; confirm with pyrometry later
Camber in roll (Roll analysis)	Loaded outer $\approx -0.5^\circ$ to -1.5° at max roll	Outer tyre near-flat under roll	Shorten/angle upper arm (more gain) — but watch RC migration
Camber gain (Bump analysis)	~ 0.5 – 1.5° neg per 25 mm bump	Smooth, monotonic	Adjust upper/lower arm length ratio & inclination
Front roll-centre height (Roll)	~ 0 – 50 mm above ground	In band, both ends	Raise/lower inner pickups together to move IC
Rear RC height	Slightly higher than front	Rear $\geq$ front (tilts roll axis)	Raise rear inner pickups (a rear-heavy car wants this)
RC migration (Roll, full travel)	Small — keep lateral/vertical move low	Balance stable through roll	Lengthen arms (less migration) — trades against camber gain
Bump steer (Steer+Bump)	$\leq \sim 0.05^\circ$ toe per mm (≈ 0)	$< \sim 0.3$ – 0.5° over full travel	Move tie-rod INNER height/length onto the arc; iterate
Caster	3 – 7°	In band	Tilt steering axis (upper vs lower outer pickup fore/aft)

Check (and which analysis)	Target (starting)	Pass criteria	If outside target → move this
Scrub radius	Small, $ \text{value} \leq \sim 25\text{--}30\text{ mm}$	Minimal kickback	Adjust wheel offset / lower outer point / KPI
KPI/SAI	As low as packaging allows	Just enough for self-centring	Move outer pickups; don't over-rely on KPI for camber
Ackermann	Per tyre peak-slip study	Matches tyre needs at op. load	Move steering-arm length / rack position; tie to Book One
Anti-dive / anti-squat	0–30%	Modest; platform OK	Tilt side-view arm angles; don't over-do (harshness)
Motion ratio (Bump)	$\sim 0.5\text{--}1.0$, near-linear	$< \sim 10\%$ variation over travel	Re-position pushrod/rocker geometry; re-check in Phase 5

The camber-gain vs roll-centre conflict (expect it)

You will find you cannot maximise camber gain AND minimise roll-centre migration — short arms give gain but a migrating RC; long arms give a stable RC but little gain ([Book Two Part E](#)). Resolve it the usual way: prioritise a STABLE roll centre and accept moderate bump camber, because a migrating RC changes balance unpredictably mid-corner, which is harder to drive than slightly imperfect camber. Document the trade — it is a design-event question.

Step 3.3 — Check travel, clearances and steering effort

1. **Travel** — confirm the geometry delivers at least the rule-minimum usable wheel travel (e.g. $\pm 25.4\text{ mm}$) plus margin, without the arms binding or the tie-rod hitting anything.
2. **Clearances** — with the model at full bump/droop and full steer, check tyre-to-upright, tyre-to-chassis, and arm-to-arm clearances in CAD. Clashes found now are free; found at assembly they are expensive.
3. **Steering** — estimate steering effort and ratio from caster/trail + tyre Mz ([Book One](#)); confirm the rack ratio keeps the tyre in its peak-slip band and the effort is comfortable for a 5th-percentile driver.

Step 3.4 — Steering geometry (designed WITH the suspension, not after)

Steering is part of THIS freeze, because its outer points live on the upright and its tie-rod shares the suspension's motion. Design it now, driven by the tyre data from [Phase 1 \(Book One\)](#).

1. **Rack choice & position** — pick and position the steering rack so its ratio keeps the tyre in its peak-slip band at the steer angles your tightest corner needs, and its travel reaches full lock without over-travel. The rack's height and length set where the tie-rod inner point must go.
2. **Steering arm & Ackermann** — set the steering-arm length and tie-rod outer point for the Ackermann % your tyre wants, balancing low-speed scrub against high-speed stability.
3. **Bump steer** — now match the tie-rod INNER point to the suspension arc so toe-change through travel is $\sim$ zero. (This is why the rack must go in before bump steer can be solved — Table D-0.)
4. **Column & effort** — estimate steering effort from caster/trail $\times$ tyre Mz ([Book One](#)); confirm a 5th-percentile driver can turn it; lay out the column, U-joints and quick-release to clear the driver and chassis.

Table 3-2. Steering geometry checks — frozen in [Phase 3](#) alongside the suspension hardpoints (Table 3-1), because the two share the upright and the tie-rod arc.

Steering check (analysis)	Target (starting)	Pass criteria	If outside target → do this
Steering ratio	Keeps tyre in peak-slip band at max steer	Comfortable lock-to-lock; tyre not past peak	Change rack ratio / steering-arm length
Ackermann	Per tyre peak-slip vs load (Book One)	Matches tyre at operating load	Move steering-arm length / tie-rod outer point
Bump steer (Steer+Bump analysis)	$\leq \sim 0.05^\circ$ toe per mm (≈ 0)	$< \sim 0.3\text{--}0.5^\circ$ over full travel	Adjust tie-rod inner height/length onto the arc
Steering effort	Comfortable for 5th-%ile driver	Not fatiguing at low speed	Reduce caster/trail a touch; change ratio
Column / U-joint phasing	Small, equal joint angles	Smooth torque, no bind, clears driver	Re-route column; add/space U-joints

❏ Order now: wheels, steering rack, rod ends/spherical bearings

Once hardpoints are converging, the wheel offset, steering rack travel/ratio, and the rod-end/spherical-bearing sizes are defined — order them (medium lead time). They must be in hand before the uprights and arms are finalised in **Phase 4**, because the bearing bores and rod-end threads set the part interfaces.

🔒 GATE G3 — hardpoint freeze (the big one)

Pass when EVERY row of Table 3-1 is within target simultaneously, travel and clearances pass, steering is confirmed, and the camber/RC trade is documented. Then FREEZE the hardpoint coordinates and issue them to the chassis and structures teams. After this gate, a hardpoint changes only through a logged change-review with sign-off — because the chassis tabs, uprights and rockers are now being designed to these exact points. An unfrozen **Phase 3** is the number-one reason FSAE cars are late or ill-handling.

Phase 4 — Load cases & component design (Weeks 9–16)

With geometry frozen, design the physical parts — uprights, hubs, bearings, A-arms, pushrods, rockers, ARBs, steering — and prove each with FEA against realistic load cases. This phase also triggers the big procurement. It overlaps **Phase 3**'s tail: start load cases as soon as hardpoints converge.

Step 4.1 — Build the load cases

Compute the forces the tyres put into the suspension in the worst moments, then multiply by a dynamic magnification factor (DMF) to cover transient/impact loading (standard practice).

Table 4-1. Suspension load cases. Build them from YOUR tyre μ and load transfer (Books 1–2), not a guessed g-number. Resolve each into forces at every pickup.

Load case	How to size it	Typical DMF	Governs
Max lateral (cornering)	$\mu \times$ loaded Fz on outer tyre (from tyre data + load transfer)	~1.3–1.5	Upright, arms, bearings
Max braking	Brake force + forward load transfer	~1.3–1.5	Upright, lower arm, chassis tabs
Combined (corner+brake)	Friction-ellipse combination (Book One Part VII)	~1.3–1.5	Worst-case upright/arm
Max longitudinal (accel)	Traction force + rear load transfer	~1.3–1.5	Rear upright, driveshaft, arms
Bump/kerb strike	Vertical, $2\text{--}3 \times$ static wheel load	2–3	Pushrod, rocker, bearing, arm
Combined worst case	Lateral + vertical + braking together	~1.5	Final sizing of critical parts

Step 4.2 — Design and FEA each component

Design in CAD, then run FEA (ANSYS/SolidWorks Simulation) under each load case. Check BOTH strength and stiffness — a part can be strong but too flexible, and compliance quietly undoes your geometry (**Book Two**).

Table 4-2. Component FEA targets. FoS on YIELD for ductile metals; add fatigue margin for endurance-critical parts. Stiffness matters as much as strength.

Component	Material (typical)	FEA target	If it fails → do this
Upright	7075-T6 Al (or Ti/steel)	Yield FoS ≥ 1.5 ; low deflection (camber compliance $< \sim 0.1\text{--}0.2^\circ$ at max lateral)	Add material at the hotspot; increase section; rib it
Wheel hub	7075-T6 / steel	FoS ≥ 1.5 ; bearing seats true	Increase diameter/wall; check bearing fit
Wheel bearings	Two tapered rollers / angular-contact	L10 life $\geq$ season; takes combined radial+axial	Up-size; preload correctly (your single-ball fix was right)
A-arms	4130 steel tube / Al	FoS ≥ 1.5 ; buckling FoS $\geq 2\text{--}3$ on compression members	Bigger tube OD/wall; better node triangulation
Pushrod	4130 / Al / CF	Buckling FoS $\geq 2\text{--}3$ (it's a strut!)	Increase diameter; shorten unsupported length

Component	Material (typical)	FEA target	If it fails → do this
Rocker	7075-T6 Al	FoS ≥ 1.5 ; stiff pivot; low deflection	Thicken; add bearing support; reduce lever arm
ARB + blades	Spring steel	Stress < limit at max roll; adjustable range OK	Resize bar dia / blade length for stiffness range
Steering arm/tie-rod	Steel/Al	FoS ≥ 1.5 ; no buckling in tie-rod	Up-size; check thread/rod-end rating

Design-for-manufacture & FMEA before you freeze

For every part ask: can WE actually make this (CNC axes, welding access, tube bends)? Can it be inspected and replaced between sessions? Run an FMEA on single-failure-point parts (upright, hub, pushrod, steering) — a suspension failure in endurance ends your competition. Load rod-ends and spherical bearings AXIALLY, never in bending (**Book Two Part N**). Design in the DAQ mounts now (shock-pot tabs, strain-gauge access on pushrods).

Step 4.3 — Steering components (freeze with the uprights)

Design and FEA the steering parts to the same standard as the suspension, from the Phase-3 geometry. The steering arm is part of the upright, so they are drawn and frozen together.

Table 4-3. Steering component checks. Rod ends and tie-rods carry AXIAL load only — never bending (**Book Two Part N**).

Component	Material (typical)	FEA / check target	If it fails → do this
Steering rack / mount	Al or steel	Mount stiff; no flex under tie-rod load	Stiffen/brace the mount to the chassis
Tie-rods	Steel/Al + rod ends	FoS ≥ 1.5 ; no buckling; axial load only	Up-size; shorten; higher-rated rod ends
Steering arm (on upright)	7075-T6 / steel	FoS ≥ 1.5 ; stiff (sets Ackermann accuracy)	Thicken; shorten lever; rib it
Column + U-joints	Steel	Takes driver torque; small joint angles	Up-size shaft; re-phase/ space joints
Quick-release + wheel	Per rules	Secure, rules-legal, clears legs	Compliant QR; reposition wheel

Procure — order the rack (if not already from **Phase 3**), column, U-joints, quick-release and steering wheel now — medium lead, and the quick-release and wheel are scrutineering items, so do not leave them to the end.

❑ MAJOR procurement window — order long-lead items NOW

This is the big order. Place them as soon as each design is FEA-passed, before the full design freeze if lead time demands: DAMPERS (longest lead — DNM/Öhlins/Fox/MCS/Penske; your DNM RCP2S), billet 7075 and 4130 tube, wheel bearings, rod ends/uniballs, fasteners (aerospace-grade — Carroll Smith), and DAQ sensors (shock pots, IMU, strain gauges) so the harness is ready for the build. Factor Bangladesh import/customs time into every order-by date. Update the procurement tracker.

🔒 GATE G4 — design freeze

Pass when: every component passes its FEA targets (Table 4-2) under all load cases; DFM and FMEA are done; drawings and the BOM are released; and long-lead parts are ORDERED with confirmed delivery dates that fit the build timeline. After G4, part geometry changes only through change-review.

No cutting material until this gate is signed.

Phase 5 — Springs, dampers & ARB sizing (Weeks 12–17)

Geometry and parts are frozen; now pick the rates. These you CAN change at the track, so set sensible, order with adjustment range, and plan to fine-tune in **Phase 8**. Overlaps **Phase 4** — you need the motion ratio from the frozen rocker.

Step-by-step with the maths (Book 2 Parts F–G)

1. **Spring rate** — use the **Phase 2** ride-frequency target and the frozen motion ratio. Wheel rate $K_{\text{wheel}} = (2\pi f)^2 \times m_{\text{sprung,corner}}$; spring rate $K_{\text{spring}} = K_{\text{wheel}} / MR^2$. Pick the nearest available coil.
2. **ARB** — from the roll-gradient target and the springs' roll contribution, size the ARB to make up the difference, and set its blade/adjustment range to cover the TLLTD span you want (front and rear bars).
3. **Dampers** — select dampers whose adjustable range brackets your target damping ratio ($\zeta \approx 0.6\text{--}0.7$), with independent bump/rebound and ideally low/high-speed. Specify an initial valving; refine with histograms in **Phase 8**.

Table 5-1. Spring/damper/ARB sizing checks. Always convert through the MEASURED motion ratio once the car exists (**Phase 7**).

Check	Target (starting)	Pass criteria	If outside target → do this
Ride frequency (from chosen spring)	2.5–3.5 Hz; F slightly < R	Within band after rounding to real spring	Pick next available rate; re-check with real MR
Wheel travel used	$\geq$ rule min + margin at max bump	No bottoming at max load case	Softer spring / more travel / bump stop tune
Roll gradient (spring+ARB)	1.0–2.0 deg/g	In band at max lateral g	Resize ARB; do NOT just stiffen springs
TLLTD adjust range	Covers $\pm$ a few % about target	Bars can trim balance without changing ride	Add blade adjustment / different bar dia
Damping ratio range	Brackets $\zeta \approx 0.6\text{--}0.7$	Adjusters reach both sides of target	Different damper spec/valving

❑ Order springs & ARB stock

Order a small RANGE of spring rates around your calculated value (one or two steps each side) — you WILL want to trim ride frequency and balance on track, and spring lead time should not stall testing. Order ARB bar stock/blades. Dampers were ordered in **Phase 4** (longest lead).

🔑 GATE G5 — rates frozen for first build

Pass when spring rates, ARB sizes/ranges and damper spec are chosen from the targets and ORDERED (with a range of springs). These are your Phase-7 starting setup; they are expected to change on track, so “frozen” here means “good enough to build and baseline,” not “final.”

Phase 6 — Fabricate, assemble & QC (Weeks 15–24)

Turn drawings into a rolling chassis. The theme of this phase is: build to the frozen numbers, and PROVE you did with inspection. A beautifully analysed car assembled 5 mm off its hardpoints handles like a different, un-analysed car.

Step-by-step

1. **Jig** — build a welding jig/fixture that holds every suspension pickup at its frozen coordinate while you weld chassis tabs and A-arms. The jig is what makes the real car match the model — do not skip it.
2. **Make the parts** — machine uprights, hubs, rockers (CNC/lathe per drawings); weld A-arms and pushrods with proper fixturing and the right filler; ream/press bearing seats to tolerance.
3. **Assemble** — assemble each corner: bearings with correct PRELOAD, fasteners to spec TORQUE (and mark them), rod ends with adequate thread engagement and in pure axial loading.
4. **DAQ** — build the DAQ harness in parallel (**Book Two** Parts J–K): shock pots, IMU near CoG, wheel-speed rings, steering & brake-pressure sensors, strain-gauged pushrods if used; bench-test every channel before it goes on the car.

Quality control — prove the build matches the design

Table 6-1. Build QC. This table is how you earn the right to trust every number you measure in *Phase 8*.

QC check	How	Pass criteria	If it fails →
Hardpoint positions	Measure pickups vs CAD (CMM, FARO arm, or careful jig/measure)	Within a few mm of frozen coords	Shim/re-tab; re-check bump steer & RC after
Wheel alignment	Camber/toe/caster gauges on setup pad	Matches design static values	Adjust shims/turnbuckles to target
Bearing preload	Torque/rotation check	Per spec; smooth, no play	Re-shim/adjust nut; recheck
Fastener torque	Torque wrench + witness mark	All to spec, all marked	Re-torque; log
Corner function	Cycle suspension by hand	Full travel, no bind, no clash	Find & clear the interference
DAQ channels	Bench + on-car power-up	Every channel reads sane & calibrated	Re-wire/re-calibrate (Book Two Part K)

GATE G6 — rolling chassis

Pass when all corners are assembled to spec, QC Table 6-1 passes (hardpoints verified against CAD!), the steering works end-to-end with correct free play, and the DAQ powers up with every channel reading sanely. The car rolls, steers and is instrumented. Now you can measure it.

Phase 7 — Shakedown & baseline setup (Weeks 23–26)

Before chasing lap time, establish the car's real, measured baseline and prove it runs reliably. Measure first, drive gently second, tune later.

Step 7.1 — Measure the real car (validate the model inputs)

Table 7-1. Baseline measurements. Each either validates a model input or sets a baseline setup value. Feed every real number back into the sim.

Measurement	Method	Validates / sets	If off from design →
Corner weights & cross-weight	Four scales, driver aboard, target fuel	Weight distribution; set cross $\approx 50\%$	Adjust perch heights; revisit ballast/packaging
CoG height	Tilt test (Book Two Part B)	Load-transfer model	Update sim with real value; if high, investigate
Motion ratio	Move wheel known amount, read shock pot	Wheel rate, ride freq, damping	Recompute rates with real MR; adjust springs if needed
Bump steer	Fix steering, cycle travel, measure toe	Confirms Phase 3 geometry as-built	If present, shim tie-rod inner to the arc
Static alignment	Camber/toe/caster gauges	Set baseline to design targets	Dial to targets; record on setup sheet
Cold/hot tyre pressures	Gauge + run	Baseline pressures (Book One)	Set cold to hit target hot

Step 7.2 — Shakedown drive (reliability before performance)

1. **First roll** — on a safe, open area, do low-speed functional checks: steering sense & return, brakes, no fluid leaks, no fouling at full steer/bump, DAQ logging cleanly.
2. **Build up** — build up gently over several short runs; after each, inspect fasteners (witness marks), rod ends, welds, tyre temps, and the data log. Fix every niggle before adding speed.
3. **Confirm** — once reliable at moderate pace, do a few harder laps to confirm nothing loosens, overheats or cracks. This is also your first real data for Phase 8.

GATE G7 — reliable runner

Pass when: the car completes repeated runs with no failures, no leaks, no loosening (marks intact), full travel without fouling, and clean data on every channel; corner weights, CoG, motion ratio and bump steer are measured and fed back into the sim; and a documented baseline setup exists. Now — and only now — chase performance.

Phase 8 — Test & validate (Weeks 25–32)

Now you turn the baseline car into a fast, correlated one. Every test below follows the same loop: run a defined test, read a specific channel, compare to target, apply the corrective action, re-run. This is the heart of “run this test, read the value, correct that” — and it is exactly the correlation loop from **Book Two Part L**. Change ONE thing at a time and log everything.

Step 8.1 — The validation test program

Table 8-1. The validation program. Run top-to-bottom; each row is a measure→compare→correct loop. Re-baseline after any change that went backwards.

Test	You measure	Target / what good looks like	If off → corrective action
Constant-radius skidpad	Understeer gradient (steer vs lat-g); peak lat-g	Mild stable understeer; lat-g $\approx$ tyre μ prediction	Push mid-corner: soften front ARB / stiffen rear. Loose: opposite. (Book Two Part M)
Skidpad μ correlation	Measured lat-g vs sim prediction	Within a few % of sim	Back out true $\lambda\mu$ (grip scaling) and update the tyre model (Book One)
Step-steer	Yaw-rate response & settling time	Quick, well-damped, no overshoot	Sluggish → less understeer/less yaw inertia. Twitchy → toward understeer/more rear
Straight-line braking	Decel g; front/rear lock order; bias	Threshold without early lock; target decel	Rear locks first → bias forward. Fronts lock early → bias rearward
Damper histogram runs	Damper-velocity histogram per corner	Smooth bell shape, centred near zero	Spike at zero → less low-speed. Long high-speed tails → more high-speed (Book Two Part G)
Contact-patch variation	Pushrod-load (or spring×pot) RMS over bumps	As low as practical	High → tune high-speed damping; confirm unsprung-mass benefit
Entry/exit balance	Driver + brake/throttle/steer traces	Balanced on entry and on power	Entry: dampers & brake bias. Exit: diff & rear damping (Book Two Part M)
G-G diagram	Lateral vs longitudinal g scatter	Fills the ellipse; edges reached	Gaps → driver coaching or setup limiting that zone (Book One Part VII)
Endurance simulation	Temps, pressures, lap consistency, wear over ~22 km	No overheating, stable pace, no failures	Fading grip → pressures/compound/cooling; any failure → back to design review

Step 8.2 — Close the correlation loop

For every test, record MEASURED vs PREDICTED. Each comparison either validates your model (now trusted for next year) or exposes an error (fix it). Priorities to correlate: roll gradient, understeer gradient, skidpad μ , and contact-patch variation. A team that closes even three of these is operating at a different level (**Book Two Part L**).

Tuning discipline (this is where seasons are won or lost)

One change at a time. Same driver, same corner, back-to-back runs. Judge with the DATA and the driver's words together. Log every change on the setup sheet with the result. Keep a known-good baseline you can always return to. Teams that “change three things and chase a feeling” never converge; teams that run this loop methodically arrive at competition with a fast, understood car.

** GATE G8 — performance signed off**

Pass when: measured handling meets the Phase-0/2 targets (lateral g, balance, braking); the model is correlated on the priority channels; the car completes an endurance-distance run with stable pace and no suspension failures; and a validated competition setup (plus a wet-weather variant) is recorded. The car is now fast AND understood.

Phase 9 — Competition readiness & scrutineering (Weeks 30–34)

A fast car that fails scrutineering scores zero. This phase makes the car legal, presentable and supportable, and prepares the people and spares.

Step 9.1 — Pre-scrutineering self-check (suspension & steering)

Table 9-1. Suspension/steering scrutineering self-check. Do this BEFORE you travel — fixing it at the event costs you track time. Verify every item against the current rulebook.

Scrutineering item	Check yourself first	If it fails →
Wheel travel	$\geq$ rule minimum usable, with driver, demonstrable	Adjust spring/geometry/bump-stops to show travel
Functional dampers	Visibly working front & rear	Fix/replace; no solid-mounted corners
Steering free play / stops	Within limit; positive lock stops fitted	Add/adjust stops; remove slack in joints
Wheel fasteners	Correct retention per rules	Fit specified retention
Tilt test (60°)	No fluid leak; no rollover (CoG/track check)	Lower CoG / widen track / fix leaks
Ground clearance	Nothing but tyres touches in events	Raise ride height / undertray
Ball-joint/rod-end retention	Captive per rules; no single-shear failure risk	Add retainers; redesign if needed

Step 9.2 — Make the car supportable at the event

1. **Setup sheets** — one per dynamic event (skidpad, autocross, endurance) plus a wet setup, each a known, tested configuration — not a guess made in the paddock.
2. **Spares & tools** — spares for every single-failure-point part (upright, hub, pushrod, rod ends, a spring set, a damper), plus alignment tools, scales, pyrometer, pressure gauge, torque wrenches, and the DAQ laptop.
3. **Drivers & crew** — drivers have seat time in the final car; everyone knows the setup-change procedure and the between-runs checklist (fasteners, temps, pressures, data).

GATE G9 — competition ready

Pass when: the car passes your own full scrutineering self-check against the current rulebook; event setup sheets exist and are tested; spares, tools and the DAQ kit are packed; drivers are trained; and the between-runs checklist is in hand. You now have a competition-ready, validated Formula Student suspension — which was the goal of all three books.

Appendix A — Decision-gate summary (print this)

Appendix A. The ten gates. Never carry an open problem across a gate.

Gate	You may NOT proceed until...
G0 Targets	Rules extracted, performance targets set, lock-list (Table 0-3) filled & signed
G1 Tyre	Tyre chosen with data; validated model; operating-load metrics recorded
G2 Concept	Vehicle target sheet frozen (WB, track, mass, CoG, freqs, roll gradient, architecture)
G3 Hardpoints	All Table 3-1 targets met together; travel/clearances OK; coordinates FROZEN
G4 Design	All parts pass FEA; DFM/FMEA done; drawings & BOM released; long-lead parts ORDERED
G5 Rates	Springs/ARB/dampers sized & ordered (with a spring range)
G6 Rolling chassis	Built to spec; QC passed (hardpoints verified!); DAQ live
G7 Runner	Reliable repeated runs; car measured; baseline setup documented
G8 Performance	Targets met; model correlated; endurance-distance run clean; setups recorded
G9 Competition	Scrutineering self-check passed; setups/spares/drivers ready

Appendix B — Corner setup sheet (template)

Appendix B. Copy per run. Also record: date, track, ambient/track temp, fuel, driver, tyre set & heat cycles, and the ONE change vs last run + its result.

Parameter	Front L	Front R	Rear L	Rear R
Cold pressure				
Hot pressure				
Camber				
Toe				
Caster			—	—
Spring rate				
Ride height				
ARB setting				
Damper bump				
Damper rebound				
Corner weight				

Appendix C — Between-runs & test-day checklist

- Fasteners: witness marks intact (walk every suspension/steering fastener)
- Rod ends / ball joints: no play, retainers in place
- Welds & tubes: no cracks, no paint flaking at joints
- Tyres: pressures (hot), temps across tread (I/M/O), wear, heat-cycle count
- Dampers: no leaks; adjusters at recorded clicks
- Ride heights / corner weights unchanged (or re-set)
- DAQ: all channels logged this run; download & back up
- Log the ONE change made and the measured + felt result

Appendix D — How the three books fit the process

Appendix D. The trilogy as one workflow: why (Book One) → how it couples (Book Two) → do this, then this (Book Three).

When you're doing...	Open this book
Choosing/modelling the tyre (Phase 1)	Book One — Tyre Field Guide (Parts XI–XIV)
Setting concept targets, balance, load transfer (Phase 2)	Book Two — VD Handbook (Parts A–D)
Designing geometry (Phase 3)	Book Two — Part E (kinematics)
Sizing springs/dampers/ARB (Phase 5)	Book Two — Parts F–G
Understanding couplings before any change	Book Two — Part H (coupling map)
Building telemetry (Phase 6)	Book Two — Parts J–K
Reading data / correcting setup (Phases 7–8)	Book Two — Parts L–M; Book One Part VII (g-g)
Executing the whole build	Book Three — this playbook

BOOK FOUR

| FORMULA STUDENT VEHICLE DYNAMICS

Practical & Reference

Fabrication and workshop skills, project management and budget, competition-day operations, and the templates pack.

Fabrication & Workshop Skills

Analysis designs the car; the workshop decides whether the real car matches the analysis. This chapter is the hands-on knowledge a VD team needs to turn drawings into trustworthy hardware. None of your clever geometry survives a bad weld, a loose bearing, or a part built 5 mm off its hardpoints.

TIG welding (the FSAE default)

Tungsten inert-gas (TIG) welding is the precise, clean process used for suspension steelwork because it gives fine control and strong, tidy joints. Key practice:

- **Material** — AISI 4130 chromoly is stronger than mild steel but more sensitive — use the right filler (e.g. ER70S-2), keep heat input controlled, and do not quench. For highly stressed 4130, consider normalising after welding.
- **Prep** — clean to bright metal, tight fit-up, proper tube coping (fish-mouth) so joints seat fully. Gaps = weak, distorted welds.
- **Distortion** — tack in sequence to balance shrinkage; weld in a jig/fixture so the frame cools to the DE-SIGNED coordinates, not wherever it pulls. This is how the built car matches CAD.
- **HAZ** — the heat-affected zone (HAZ) beside a weld is weaker — design joints so the HAZ is not at a peak-stress point, and never weld right next to a bearing bore (heat ruins the fit).

DO	DON'T
✓ Weld in a fixture that holds hardpoints to coordinate.	✗ Don't weld near bearing seats or threaded inserts.
✓ Coping/fit-up tight before striking an arc.	✗ Don't force gaps closed with the weld puddle.
✓ Use the correct filler and controlled heat for 4130.	✗ Don't quench chromoly to "save time".
✓ Inspect every weld (visual, and dye-penetrant on critical joints).	✗ Don't grind welds flush on thin tube (removes section).
✓ Let assemblies cool in the jig before releasing.	✗ Don't trust a weld you cannot see the back of — design for access.

Machining & fits

Uprights, rockers and hubs are machined; their accuracy sets your geometry and bearing life.

- **Process** — CNC-mill uprights, rockers and bell cranks from 7075-T6; lathe the hubs. Hold tight tolerance on bearing bores and the pickup-point spacing.
- **Bearing fits** — use the correct interference/clearance for each bearing (e.g. an H7 bore for a press fit); a loose bore spins the outer race and destroys the upright, a tight one crushes the bearing. Follow the bearing maker's fit table.
- **Finish** — deburr every edge and radius internal corners — sharp corners are fatigue crack starters on a part that sees millions of cycles.

Composites (undertray, seat, bodywork)

- **Method** — wet layup (hand + resin) is cheapest and what most FSAE teams use for an undertray/seat; prepreg + oven is lighter and more consistent but needs an autoclave/oven. Vacuum-bagging a wet layup is a cheap way to improve consolidation.
- **Quality** — mould prep (release agent), correct ply orientation for the loads, full wet-out, and a proper cure schedule decide the part. Post-cure trim with the right PPE.

Composites safety is not optional

Carbon/glass dust and uncured resins are hazards. Wear a respirator (not a dust mask) when sanding/trimming composites, nitrile gloves with resin, and eye protection. Work in ventilation. A fast lap is not worth a lung or a sensitised skin reaction — this is a hard team rule, not a suggestion.

Fasteners, rod ends & torque

Suspension is held together by threaded joints that see full load and vibration. Treat fasteners as engineered components, not hardware-store bolts (Carroll Smith's books are the classic reference here).

Practice	Why
Use proper-grade bolts (e.g. aerospace AN/MS or ISO 10.9/12.9)	Suspension loads need rated fasteners, not mystery steel
Adequate thread engagement ($\approx 1 \times$ dia in steel, $\approx 2 \times$ in aluminium)	Too few threads strips under load
Locking: nyloc, castellated + split pin, or thread-locker per joint	Vibration backs off plain nuts
Torque to spec and witness-mark every fastener	Set correct preload; spot a backed-off bolt at a glance
Rod ends / spherical bearings loaded AXIALLY only	They are weak in bending — bending is how they fail
Jam nuts on adjustable rod ends; check thread showing	Locks the length; confirms engagement
Anti-seize on Ti/stainless and dissimilar metals	Prevents galling and galvanic seizing

DO	DON'T
✓ Torque-wrench and witness-mark every suspension fastener.	✗ Don't reuse nylocs many times — they lose grip.
✓ Keep a fastener spec on the drawing (grade, torque, locking).	✗ Don't mix metric/imperial threads.
✓ Re-check torque after the first few runs (bedding in).	✗ Don't over-torque aluminium — you strip it.
✓ Load rod-ends in pure tension/compression.	✗ Don't leave a rod end in single shear where it can bend.

Project Management, Roles & Budget

A Formula Student car is as much a project-management challenge as an engineering one. Most teams that fail to finish fail on schedule, money or communication — not on physics. This chapter keeps the VD sub-team on track.

Own the schedule

Build a season Gantt from the ten-phase timeline (Book Three) and protect the freeze dates. Every week a hardpoint stays “open” is a week chassis and uprights cannot finalise. Track critical-path items (dampers, imports) weekly.

Roles (a small-team RACI)

Task area	Responsible	Accountable	Consulted
Tyre data & model	Tyre lead	VD lead	Sim owner
Kinematics & hardpoints	VD lead	VD lead	Chassis, structures
Steering	Steering/VD	VD lead	Ergonomics, structures
Component design & FEA	Structures owner	VD lead	Manufacturing
Springs/dampers/ARB	VD lead	VD lead	Sim owner
Fabrication	Manufacturing owner	Team lead	VD, structures
DAQ & testing	DAQ owner	VD lead	Whole team
Procurement & budget	Procurement owner	Team lead	All sub-leads

Budget, procurement & customs

- **BOM discipline** — build a BOM with specs, supplier, quote, lead time, order-by date, and cost; keep it live. The BOM is your budget and your schedule in one sheet.
- **Lead time & customs** — for a Bangladesh-based team competing at FSCC, import and customs time is real and variable — add generous margin to the order-by dates of anything shipped in (dampers, bearings, tyres, TTC-driven parts). Line up suppliers and payment early (**Phase 0**).
- **Budget tracking** — track committed vs spent vs remaining; keep a contingency ($\approx 10\text{--}15\%$) for breakages and re-makes, because you WILL break a part in testing.

Change control after a freeze

After a gate freeze (hardpoints at G3, parts at G4), a change is not a quiet edit — it is a logged change-review: who asked, what moves, what it affects (use the coupling map, **Book Two Part H**), who re-checks, and sign-off. This one habit prevents the classic disaster where someone “just nudged” a pickup point and three finished parts no longer fit.

Competition-Day Operations

A fast, validated car still has to pass inspection, survive the paddock, and be supported run-to-run. Points are lost here that no amount of design can recover.

Before you travel

- **Spares** — spares for every single-failure-point part (upright, hub, pushrod, rod ends, a spring set, a damper), fasteners, and consumables.
- **Tools** — alignment gauges, corner scales, pyrometer, pressure gauge, torque wrenches, the DAQ laptop and cables, and a printed setup book.
- **Setups** — tested setup sheets for each dynamic event plus a wet setup — decided and logged in advance, not guessed in the paddock.

Scrutineering — suspension & steering self-check

Run this BEFORE you travel; fixing it at the event costs track time. Verify every item against the current FSCC/FSAE rulebook — numbers change year to year.

Item	Check	If it fails
Wheel travel	$\geq$ rule minimum usable, with driver, demonstrable	Adjust spring/geometry/bump-stops
Functional dampers	Visibly working front & rear	Fix/replace; no solid corners
Steering	Free play within limit; positive lock stops	Add/adjust stops; remove joint slack
Wheel & fastener retention	Per rules	Fit specified retention
Tilt test	No fluid leak and no rollover at the required angle	Lower CoG / widen track / fix leaks
Ground clearance	Only tyres touch during events	Raise ride height / undertray
Joint retention	Rod ends/ball joints captive, no single-shear risk	Add retainers; redesign if needed

In the paddock & between runs

- **Setup rhythm** — corner-weight on arrival (travel shifts things), set tyre pressures cold to hit target hot, confirm alignment, zero the DAQ.
- **Between-runs** — walk every suspension/steering fastener (witness marks), check rod ends and welds, read tyre temps/pressures, download and back up data. One change at a time, logged.
- **Driver & comms** — clear driver hand signals and radio calls; a plan for an off, a flat, or a failure; and someone watching live telemetry for temps/pressures/voltage alarms.

Review the data every session, not just at home

The teams that improve fastest at an event run the correlation loop in the paddock: after each run, pull the log, check tyre temps, damper histograms and the g-g usage, make ONE change, and go again ([Book Two Part L](#)). An event is the richest test day you get all year — treat it like one.

Templates & Checklists

Copy these into your team drive and fill them in. They are the canonical versions; Book Three's appendices hold matching quick-reference copies.

Corner setup sheet

Parameter	Front L	Front R	Rear L	Rear R
Cold / hot pressure				
Camber				
Toe				
Caster			—	—
Spring rate				
Ride height				
ARB setting				
Damper bump / rebound				
Corner weight				

With every sheet — Also record: date, track, ambient/track temp, fuel, driver, tyre set & heat-cycle count, and the ONE change vs last run + its measured result.

Procurement tracker

Item	Spec	Supplier	Quote	Lead time	Order-by	Status	Owner

Test log

Run	Test	Setup change	Measured result	Driver note	Next action

FEA load-case sheet

Load case	Tyre forces (Fx/Fy/Fz)	DMF	Part checked	FoS / deflection	Pass?
Max lateral					
Max braking					
Combined					
Bump strike					

Load case	Tyre forces (Fx/Fy/Fz)	DMF	Part checked	FoS / deflection	Pass?
Worst case					

Design-review checklist (before each gate)

- Targets/inputs for this phase are frozen and signed
- Every number traces to data, sim, or test — not a guess
- Coupling checked: what this change also moves (Book Two Part H)
- Rules compliance verified against the current rulebook
- DFM & FMEA done for new parts
- Procurement order-by dates still fit the timeline
- Deliverables logged; gate signed with date and owner

REFERENCE

| FORMULA STUDENT VEHICLE DYNAMICS

Glossary, Symbols & Reading

One merged glossary, symbol table and annotated reading list for all four parts.

Glossary

A merged A–Z of every term used across the four parts.

Term	Meaning
4-post / 7-post rig	A lab rig that shakes the car to measure/optimize contact-patch load control.
Ackermann steering	Linkage geometry setting how much more the inside wheel steers than the outside in a corner.
Adhesion	Grip from momentary molecular bonding of rubber to the road; dominant on smooth clean surfaces.
Aligning moment (M_z)	Self-aligning torque; the twist about the vertical axis that tries to straighten the steering. Source of steering feel.
Anti-dive / anti-squat	Side-view geometry that reacts braking/acceleration load transfer through the links, reducing dive/squat.
Anti-roll bar (ARB)	A spring that resists body roll only; used to tune roll-stiffness distribution and hence balance.
Bead	Steel-wire hoop that locks the tyre to the rim and reacts loads into the wheel.
Bias-ply (cross-ply)	Construction with cords running diagonally bead-to-bead; forgiving, fast-warming; common for FSAE slicks.
BOM	Bill of materials: the full parts list with specs, suppliers, quantities and cost.
Brush model	A conceptual tyre model (tread as bristles) that explains the shape of the force curves; not fitted to data.
Bump steer	Unwanted toe change as the wheel moves in bump/roll; designed to near zero.
Camber (inclination angle, γ)	Tyre lean viewed from the front; negative = top leans in. Keeps the patch flat under roll; makes camber thrust.
Camber gain	Rate of camber change with wheel travel; set by arm lengths/angles.
Camber stiffness (C_γ)	Lateral force generated per degree of camber.
Camber thrust	Lateral force a leaned tyre makes even at zero slip angle.
Carcass (casing)	The cord-and-rubber structure that gives the tyre its shape and carries loads.
Caster	Side-view tilt of the steering axis; gives mechanical trail, self-centring and corner camber.
CoG (centre of gravity)	The point the car's mass acts through; its height drives all load transfer.
Combined slip	Simultaneous lateral and longitudinal slip; forces share one grip budget (friction ellipse).
Compliance	Unwanted flex in parts/bushings that changes geometry under load; a "leak" in your kinematics.
Composite layup	Building a part from fibre (carbon/glass) plies and resin; wet layup, prepreg, vacuum bag.
Compound	The tread rubber chemistry; sets grip level, temperature window and wear (e.g. LC0, R20).
Contact patch (footprint)	The area of tyre touching the road; area $\approx$ vertical load / contact pressure.
Contact-patch load variation	How much vertical tyre load fluctuates over bumps; lower = steadier grip.
Corner weighting	Adjusting spring perches so the diagonal (cross) weight is $\sim 50\%$.
Cornering stiffness (C_α)	Slope of lateral force vs slip angle at zero slip; N/deg or N/rad. Sets response and balance.
Critical damping	The damping that returns a mass to rest fastest without overshoot; the reference for damping ratio.
Cross weight	Diagonal weight percentage; off-50% makes the car asymmetric left vs right.
Damping ratio (ζ)	Fraction of critical damping; race cars ~ 0.6 – 0.7 for body control.
DAQ / telemetry	Data acquisition: the sensors, logging and streaming that measure the car.
DBC file	The database mapping CAN message bytes to named, scaled channels; makes logs readable.
Degradation	Loss of grip as a tyre overheats or ages (heat cycles harden the rubber).
Degrees of freedom	The six rigid-body motions: surge, sway, heave, roll, pitch, yaw.
Dynamic magnification factor	A multiplier on static loads to account for transient/impact loading in FEA.
Effective rolling radius (R_e)	Distance travelled per radian of wheel rotation; between free and loaded radius. Needed for slip ratio.
Friction circle / ellipse	The boundary of combined lateral+longitudinal force a tyre can make; one shared grip budget.

Term	Meaning
Friction coefficient (μ)	Force divided by vertical load; ~1.3–1.8 for racing slicks.
g-g diagram	Plot of combined lateral vs longitudinal acceleration the car achieves; the friction ellipse at vehicle level.
Heat cycle	One warm-up/cool-down of a tyre; cumulative cycles harden the compound and reduce grip.
Heat-affected zone (HAZ)	The weakened region beside a weld where the metal was heated but not melted.
Heave	Vertical (z) motion of the sprung mass.
Hysteresis	Grip from the road deforming the rubber as it slides; dominant on rough surfaces.
Inclination angle	Pacejka/ISO term for camber.
Instant centre (IC)	The point a wheel instantaneously pivots about, from projecting the control arms.
Jacking	The car lifting its sprung mass mid-corner due to a high roll centre; raises CoG, hurts grip.
Jig / fixture	A tool that holds parts at their exact design coordinates during welding/assembly.
KPI / SAI	Kingpin / steering-axis inclination: inward lean of the steering axis; gives self-centring and sets scrub.
Lateral force (F_y)	Cornering force; the sideways force that turns the car.
Load sensitivity	The fall of μ as vertical load rises; the reason even load sharing makes more grip than uneven.
Load transfer	Redistribution of vertical tyre load under accel/brake (longitudinal) or cornering (lateral).
Longitudinal (slip) stiffness (C_{κ})	Slope of F_x vs slip ratio at zero slip.
Longitudinal force (F_x)	Drive (+) or brake (–) force along the direction of travel.
Lumped-parameter model	Simplified whole-car model (masses, stiffnesses, tyre data) for fast handling/lap prediction.
Magic Formula (Pacejka)	Semi-empirical tyre model; coefficients B, C, D, E shape the force curve; the industry standard.
Mechanical trail	Ground lever arm from caster; adds to pneumatic trail in steering feel.
MMM (Milliken Moment Method)	Yaw-moment vs lateral-force map of car balance across the grip range; built from tyre data.
Moment of inertia	Resistance to angular acceleration; yaw inertia (I_{zz}) dominates handling feel.
Motion ratio	Spring/damper movement per unit wheel movement; sets effective wheel rate.
Motion ratio (installation ratio)	Spring/damper travel per unit wheel travel; wheel rate = spring rate $\times$ MR ² .
Neutral steer	Both axles reach the grip limit together; maximum-grip balance.
Nyquist rule	Sample at $\geq 2\times$ the highest frequency of interest (in practice 5–10 $\times$) to avoid aliasing.
Oversteer	Rear tyres reach the limit first; the car rotates/steps out.
Peak slip angle	The slip angle at which lateral force is maximum; rises with load.
Pitch	Rotation about the lateral (y) axis; dive under braking, squat under power.
Pneumatic trail (t_p)	Rearward offset of the lateral force in the patch; creates aligning moment; shrinks to ~0 at the force peak.
Preload	A deliberate clamping force set in a bearing or bolted joint to remove play and set stiffness.
Pushrod / pullrod	Link transmitting wheel load to an inboard spring/damper (rod in compression / tension).
Pyrometer	A probe that reads tyre tread temperature; used across inner/middle/outer to diagnose camber & pressure.
RACI	Responsible / Accountable / Consulted / Informed — a way to assign task ownership.
Radial	Construction with cords at 90° plus stiff belts; low rolling resistance, sharper peak; most road tyres.
Relaxation length (σ)	Distance a tyre must roll for lateral force to build after a slip change; source of response lag.
Ride frequency	Natural bounce frequency of a sprung corner; the clean way to choose springs.
Roll	Rotation about the longitudinal (x) axis in a corner.
Roll axis	The line joining front and rear roll centres; the sprung mass rolls about it.
Roll centre	The point at an axle about which the sprung mass rolls; its height splits geometric vs elastic transfer.
Roll gradient	Degrees of body roll per g of lateral acceleration (~1–2 deg/g typical).
Roll stiffness	Torque resisting roll per degree; from springs + ARBs.
Rolling resistance (M_y)	The moment resisting wheel rotation; an energy loss.
SAE axis system	Standard tyre axes: x forward, y right, z down; defines the signs of all forces and moments.

Term	Meaning
Scaling factor ($\lambda\mu$)	Multiplier that makes lab (belt) grip track-representative; a grip scaling factor $k \approx 0.65$ is typical.
Scrub radius	Distance from the steering-axis ground point to the contact-patch centre; keep small.
Scrutineering	Pre-event technical inspection that checks the car against the rules.
Shock pot	Linear potentiometer measuring damper position; the backbone VD sensor.
Slick	A treadless competition tyre; maximum dry contact area.
Slip angle (α)	Angle between where the wheel points and where it travels; the master variable for cornering force.
Slip ratio (κ)	Fractional fore-aft slip of the tread vs the road; the master variable for traction/braking.
Sprung mass	Mass supported by the springs (chassis, engine, driver...).
Static margin	A measure of how far the car is from neutral steer (fore/aft of the neutral-steer point).
Steady-state	A condition held constant (e.g. constant-radius corner) vs transient (changing).
Surge / sway	Longitudinal (x) / lateral (y) translation of the car.
Swing-arm length (FVSA/SVSA)	Front-/side-view distance to the instant centre; sets camber gain / anti-geometry.
TIG welding	Tungsten inert-gas welding; the precise, clean process used for FSAE steel/aluminium suspension.
TLLTD	Total Lateral Load Transfer Distribution: the front share of lateral load transfer; the master balance knob.
Toe	Whether wheels point in (toe-in) or out (toe-out) seen from above.
Understeer	Front tyres reach the limit first; the car runs wide.
Understeer gradient (K)	Measure of balance: $K > 0$ understeer, $K < 0$ oversteer; set by cornering stiffness and axle loads.
Unsprung mass	Mass not supported by the springs (wheel, tyre, upright, brake); low is good for grip over bumps.
Vertical load (Fz)	How hard the tyre is pressed onto the road; from weight, weight transfer and downforce.
Weight (load) transfer	Shift of vertical load between tyres under acceleration; lateral transfer loads the outside tyres.
Wheel rate	Spring stiffness measured at the wheel ($= \text{spring rate} \times MR^2$).
Yaw	Rotation about the vertical (z) axis — the car pointing into the corner.
Yaw inertia	Resistance of the car to rotating about its vertical axis; low = eager direction change.
Yaw inertia (I_{zz})	Resistance to yaw rotation; low = eager direction change.
μ_{peak}	The maximum friction coefficient a tyre reaches at a given condition.

Symbols & notation

Symbol	Quantity	Units
α	Slip angle	deg or rad
κ (SR)	Slip ratio	– (or %)
γ (IA)	Camber / inclination angle	deg
μ	Friction coefficient	–
σ	Relaxation length	m
$\lambda\mu$	Friction scaling factor	–
Ω	Wheel rotational speed	rad/s
F_x, F_y, F_z	Longitudinal, lateral, vertical force	N
M_x, M_y, M_z	Overtuning, rolling-resistance, aligning moment	N·m
C_α	Cornering stiffness	N/deg or N/rad
C_κ	Longitudinal slip stiffness	N per unit SR
C_γ	Camber stiffness	N/deg
t_p, t_m	Pneumatic trail, mechanical trail	m
R_e	Effective rolling radius	m
B, C, D, E	Magic-Formula stiffness/shape/peak/curvature factors	– (D in N)
a_y	Lateral acceleration	m/s ² or g
K	Understeer gradient	rad/g or deg/g
h_{cg}	Centre-of-gravity height	m
t	Track width	m
W	Vehicle weight	N
m	Vehicle mass	kg
L	Wheelbase	m or mm
a_x, a_y	Longitudinal, lateral acceleration	m/s ² or g
ΔF_z	Load transfer	N
I_{zz}	Yaw moment of inertia	kg·m ²
TLLTD	Total lateral load-transfer distribution (front %)	– (%)
K_{spring}	Coil spring rate	N/mm
K_{wheel}	Wheel rate (= $K_{spring} \cdot MR^2$)	N/mm
K_{roll}	Roll stiffness	N·m/deg
MR	Motion / installation ratio	–
f_{ride}	Ride frequency	Hz
φ	Body roll angle	deg
ζ	Damping ratio	–
c	Damping coefficient	N·s/m
γ	Camber / inclination angle	deg
κ	Slip ratio (Book One)	–
θ	CoG-test tilt angle	deg

Further reading

The books and resources behind this handbook, in a sensible learning order. Chapter numbers were checked against each book's table of contents.

1. **Paul Haney, The Racing & High-Performance Tire** — plain-English, no maths — construction, slip, load sensitivity, temperature, pressure. The fastest way to make the tyre part click.
2. **Milliken & Milliken, Race Car Vehicle Dynamics (SAE R-146)** — the canonical motorsport reference. Ch. 2 Tire Behavior; Ch. 5 balance; Ch. 8 Milliken Moment Method; Ch. 9 g-g; Ch. 14 Tire Data Treatment; Ch. 16–23 ride/roll rates, suspension geometry, wheel loads, steering, driving & braking, springs, dampers, compliances.
3. **Gillespie, Fundamentals of Vehicle Dynamics (SAE R-114)** — clear, concise fundamentals. Ch. 5 Ride, Ch. 6 Steady-State Cornering, Ch. 7 Suspensions, Ch. 8 Steering, Ch. 10 Tires.
4. **Pacejka, Tire and Vehicle Dynamics, 3rd ed.** — the tyre-model reference. Ch. 3 brush model, Ch. 4 the Magic Formula, Ch. 5 & 7 transient behaviour.
5. **Jorge Segers, Analysis Techniques for Racecar Data Acquisition, 2nd ed. (SAE R-408)** — the data-acquisition bible — sensors, math channels, and the exact analyses (roll gradient, damper histograms, understeer gradient, g-g) in the telemetry chapters.
6. **Allan Staniforth, Competition Car Suspension** — practical, FSAE-relevant geometry and setup — roll centres, camber, the compromises.
7. **Carroll Smith, the “to Win” series** — fasteners, materials and build craft (Engineer to Win); setup and dampers (Tune to Win); build discipline (Prepare to Win).
8. **John C. Dixon** — rigorous geometry maths (Suspension Geometry and Computation) and dampers in depth (The Shock Absorber Handbook).
9. **OptimumG (Claude Rouelle)** — free seminars and tech tips at FSAE level — load transfer, TLLTD, the Milliken Moment Method, data.
10. **Tools & data** — the FSAE Tire Test Consortium data (membership via fsaettc.org); MFeval (MATLAB File Exchange #63618) and OptimumTire to fit the Magic Formula; OptimumKinematics/Lotus for geometry; OptimumLap for lap sim.

A closing word to the next VD lead

You now hold more vehicle-dynamics knowledge than most Formula Student teams ever write down — the tyre, the car that feeds it, the plan to build it, and the craft to make it real. Team Blitz started this without seniors or faculty. This handbook is so the next person does not have to. Keep it alive: correct it with what you measure, add what you learn, and hand it on. Now go build a fast car.